Air University, Islamabad

Software Requirement Specification
(SRS DOCUMENT)

for

Project Title

Version 1.0

By

Ehtasham Ahmed 221792

Umer Shabbir 221778

Faiz ur Rehman 221788

Supervisor

Mr- Anwar ul Hassan

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Comments (by committee) ***include the ones given at scope time both in doc and presentation**	**Action Taken**

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Signature_____

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1. Introduction

The introduction of the **Software Requirements Specification (SRS)** should provide an overview of the entire **SRS**. It should include the purpose, scope, modules and overview of the **SRS**.

1.1 Purpose

The CropConnect application is a web-based decision-support system designed to alleviate uneven crop distribution across Pakistan by integrating real-time production data, market prices, weather forecasts, and transportation-network information into a single, interactive platform. Through intuitive dashboards, geospatial maps, and predictive analytics, CropConnect enables stakeholders to identify surplus and deficit regions, optimize transport routes, and monitor food-security indicators.

This SRS documents the functional and non-functional requirements of CropConnect and is intended for the following readers:

Developers: To design and implement system components in accordance with specified requirements.

Testers: To develop test cases that verify each requirement is met.

Project Managers: To track progress against deliverables and ensure that scope and quality objectives are achieved.

Operations & Maintenance Teams: To deploy, configure, and support the system in production.

End Users (Wholesalers, Distributors, Government Analysts, NGO Coordinators): To understand system capabilities, data inputs/outputs, and usage constraints.

Documentation Writers: To produce user guides, help manuals, and training materials based on these requirements.

1.2 Scope

The CropConnect is a web-based system designed to address the issue of uneven crop distribution across Pakistan by leveraging data-driven decision-making, geospatial analysis, and predictive modeling. The platform will integrate real-time data on crop production, market prices, weather conditions, and transportation infrastructure to facilitate efficient redistribution of surplus crops to regions facing shortages. Key functionalities include an interactive map displaying crop availability, demand-supply gaps, and optimized transport routes. Sellers will be able to efficiently procure crops from surplus areas to meet demand in deficit regions. Government Policy makers will have access to analytical reports to monitor food security and plan interventions accordingly.

The system will incorporate weather forecasting APIs to assess climate impacts on crop yield and market trends, considering the significant effects of recent climate change and using historical data to predict future weather changes. Predictive machine learning models will forecast market prices, helping stakeholders make informed decisions. Optimization algorithms will identify cost-effective transportation routes, reducing logistical expenses and post-harvest losses. The platform will feature a user-friendly interface, with dashboards providing visual insights through charts, graphs, and interactive reports. It will support multiple user roles (wholesalers, distributors, and policymakers etc.) ensuring a comprehensive solution to food distribution challenges. Ultimately, this project aims to enhance food security, stabilize prices, and promote sustainable agricultural practices across Pakistan.

1.3 Modules

- **Module 1: Crop Data Collection & Analysis**

  - FE 1.1 Real-Time Crop Data Collection
  - FE 1.2 Historical & Real-Time Weather Data Integration
  - FE 1.3 Demand–Supply Trend Analysis

- **Module 2: Production Forecasting**

  - FE 2.1 Seasonal Yield Prediction
  - FE 2.2 Forecast Comparison

- **Module 3: Market Intelligence & Forecasting**

  - FE 3.1 Pricing Trends Analysis
  - FE 3.2 Price & Toll Alerts

- **Module 4: Distribution Optimization & Analysis**

  - FE 4.1 Surplus/Deficit Region Identification
  - FE 4.2 Regional Comparison
  - FE 4.3 Transportation Route Planning

- **Module 5: Dashboards, Reporting & Export**

  - FE 5.1 Interactive Dashboard Visualization (Production, Pricing, Forecast, Comparison, NGO Relief)
  - FE 5.2 Export: PDF/CSV of Tables, Charts, Dashboard Snapshots
  - FE 5.3 Scheduled Email Reports

- **Module 6: Alerts & System Health**

  - FE 6.1 Central Alert Engine (ingestion failures, data-pipeline health)
  - FE 6.2 Weather Impact Alerts (ingested from weather API)
  - FE 6.3 Data Ingestion Alerts

- **Module 7: User Management & Access Control**

  - FE 7.1 Role-Based Registration (Wholesaler, Distributor, Government Analyst, NGO, etc.)
  - FE 7.2 Access Control (permissions per role, e.g., “write” vs. “read”)
  - FE 7.3 Profile & Preferences Management

1.4 Overview

This document is organized to provide a comprehensive and structured specification of the CropConnect system. It is divided into the following major sections:

Section 1: Introduction: Describes the purpose of this SRS, the scope of the CropConnect project, the high-level modules that make up the system, and this overview of the document structure.

Section 2: Overall Description: Presents a high-level view of CropConnect's context and environment, including the product perspective, the various user classes and their characteristics, the operating environments, and any design and implementation constraints.

Section 3: Requirement Identifying Technique: Explains the methodologies and techniques used to elicit, analyze, and validate the requirements presented herein.

Section 4: Functional Requirements: Details each system feature in terms of its inputs, behaviors, and outputs, organized by module (e.g., Crop Data Collection & Analysis, Production Forecasting, Distribution Optimization, and so on).

Section 5: Non-Functional Requirements: Specifies the system's quality attributes—such as reliability, usability, performance, and security—and any constraints on those attributes.

Section 6: External Interface Requirements: Defines the user interface requirements, software interfaces (APIs), hardware interfaces, and communication protocols that CropConnect must support.

Section 7: References: Lists all documents, standards, APIs, and other resources referenced in this SRS.

2. Overall Description

2.1 Product Perspective

Crop Connect is an entirely new application (Decision making tool) designed to solve the uneven Crop Distribution problem in Pakistan. It doesn't evolve from any existing system instead it uses multiple data sources & analytic services into one interface/system.

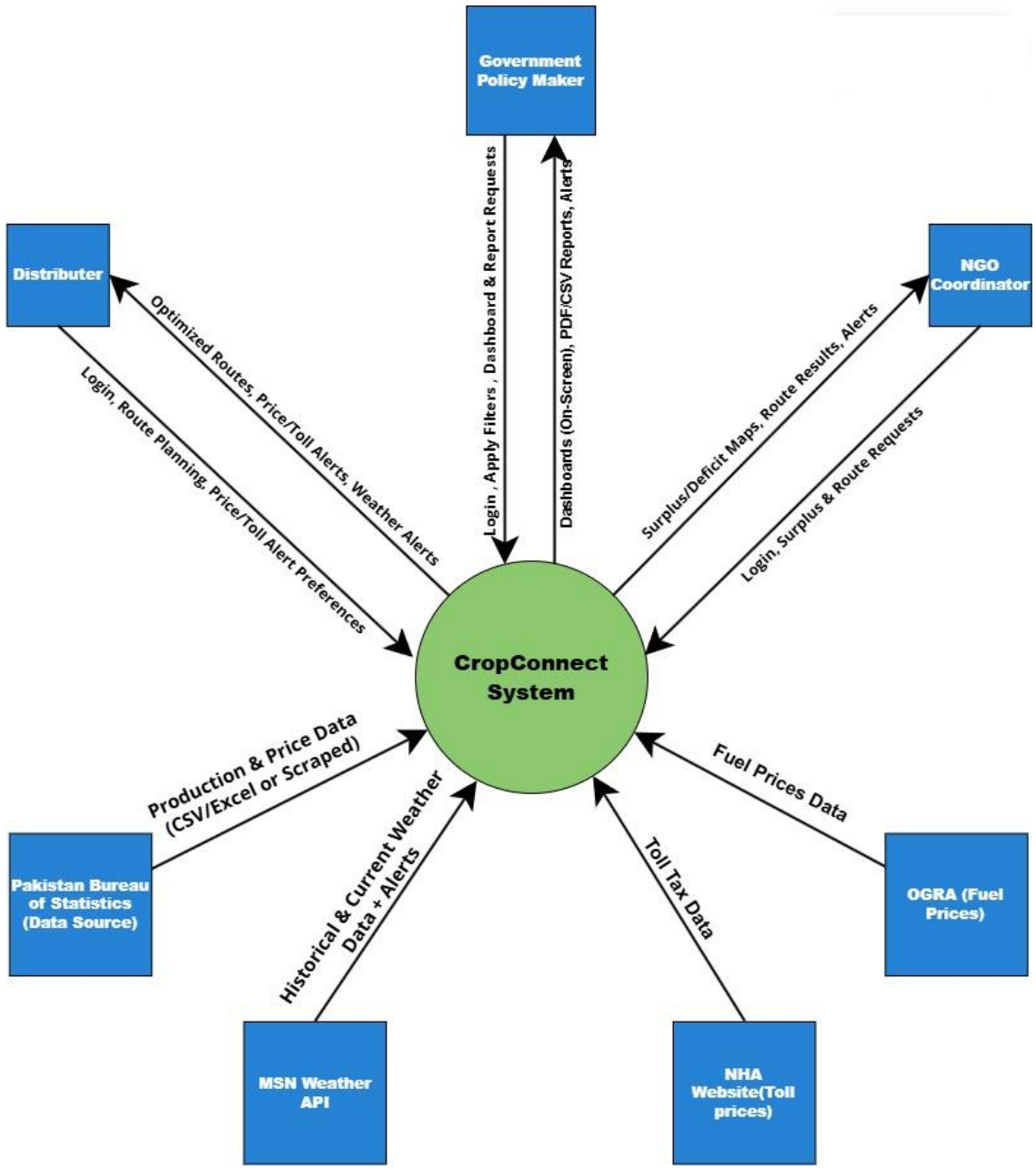


Fig A-1: Context Diagram of CropConnect System

2.2 User classes and characteristics

User Class	Description
Government Policy Maker	A government analyst who uses the system weekly to monitor crop production, pricing, and forecasts. They can generate and export PDF/CSV reports and analyze district-wise crop production trends. They will use interactive dashboards for production data analysis, pricing trends analysis, and regional analysis.
NGO Coordinator	Staff from FAO and local foundations who access the platform on an ad-hoc or emergency basis to identify surplus/deficit regions and coordinate relief shipments. They rely heavily on the surplus/deficit analysis and weather alerts for NGO operations. They can export reports to share with field teams, receive real-time alerts, and adjust schedules based on weather information.
Distributor (Logistics)	Representatives from logistics companies who use the system daily to plan transport routes and optimize costs. They will leverage route planning, weather alert monitoring, and emergency route planning. The distributor can adjust routes based on weather warnings, calculate the best cost-effective paths, and get notifications of severe weather or toll changes.

2.3 Operating Environment

OE-1: Client Platforms & Browsers

- **OE-1.1**: Supported Desktop Browsers (Windows, macOS, Linux)
  - Google Chrome: current stable release plus the two immediately preceding major versions
  - Mozilla Firefox: current stable release plus the two immediately preceding major versions
  - Microsoft Edge: current stable release plus the two immediately preceding major versions
  - Apple Safari: versions 12.0 and above (on macOS)
- **OE-1.2**: Supported Mobile/Tablet Browsers (iOS & Android WebViews)
  - Mobile Chrome (Android): current stable release plus the two immediately preceding major versions
  - Mobile Safari (iOS): iOS 12.0 and above
  - Firefox for Android: current stable release plus the two immediately preceding major versions
- **OE-1.3**: Legacy Compatibility (if required by user base)
  - Internet Explorer 11 (Windows only) for view-only access—certain interactive features (e.g., WebSockets, modern CSS grid) may degrade to fallback behaviors
  - Older Safari (versions 10.0–11.0) and Firefox (versions 50–61) are supported for basic data viewing, but advanced visualizations may not render optimally

OE-2: Client Device Requirements

- **OE-2.1**: Desktop/Laptop
  - Minimum 4 GB RAM, dual-core CPU (Intel Core i3 or equivalent)
  - Display resolution: 1280 × 720 px minimum (UI scales to 4K)
  - Network adapter: Ethernet or Wi-Fi (802.11n/ac) with a recommended minimum bandwidth of 2 Mbps
- **OE-2.2**: Tablet
  - Minimum 2 GB RAM, quad-core ARM-based processor or equivalent
  - Viewport width: 768 px to 1024 px (UI adapts with collapsible sidebar)
  - Touchscreen support for tap, swipe, and pinch gestures
- **OE-2.3**: Mobile Phones
  - Minimum 2 GB RAM, ARM-based processor (e.g., Snapdragon 625 or equivalent)
  - Viewport width below 768 px (navigation collapses into full-screen hamburger menu)
  - Network: 3G/4G/5G or Wi-Fi (802.11n/ac), with automatic fallback to low-bandwidth mode if connection < 500 Kbps

OE-3: Server & Hosting Environment

- **OE-3.1**: Backend Services
  - Deployed on a cloud provider (e.g., AWS or Google Cloud Platform) running Linux (Ubuntu 18.04 LTS or later)
  - Nodes provisioned with at least 4 vCPU and 8 GB RAM for API, analytics, and machine learning workloads
- **OE-3.2**: Database Platform
  - MongoDB, hosted on MongoDB Atlas
  - Collections will store production, pricing, forecasting, and user data in JSON format

OE-4: Network Connectivity & Security

- **OE-4.1**: Protocols & Encryption
  - All client-to-server and server-to-third-party API calls must use HTTPS (TLS 1.2 or higher)
  - WebSockets for real-time alerts shall be secured via WSS (WebSocket over TLS)
- **OE-4.2**: Bandwidth Requirements
  - Normal mode: ≥ 2 Mbps to render interactive dashboards (charts, maps, tables) without noticeable delay
  - Low-bandwidth mode: < 500 Kbps—automatic degradation to static chart images and reduced map tile resolution
- **OE-4.3**: Firewall & Port Access
  - HTTP/HTTPS (TCP ports 80/443) open to public internet for web traffic
  - Redis (TCP port 6379) and database (TCP port 5432 for PostgreSQL, 3306 for MySQL, 27017 for MongoDB) accessible only from within the same VPC or through a secure peering connection

OE-5: Geographic Considerations

- **OE-5.1**: User Access Locations
  - Users globally may access CropConnect via internet provided they can reach the cloud provider's regional endpoints in North America, Europe, or Asia
  - The web application front end is optimized for latencies up to 200 ms (client to nearest data center)
- **OE-5.2**: Server Regions for Low Latency
  - Primary application servers and databases deployed in at least two regions:
    1. **Asia (e.g., asia-south1 for India/Pakistan)**
    2. **North America (e.g., us-central1)**
  - Traffic is routed via geo-DNS so that users connect to the nearest region; automatic failover if a region becomes unavailable

OE-6: Third-Party & API Integrations

- **OE-6.1**: Weather Data API
  - MSN Weather or PBS weather endpoints must be reachable over HTTPS; expected average response < 500 ms
- **OE-6.2**: Forecasting Service
  - Machine learning model (hosted as a separate container/service) invoked via REST over HTTPS; average inference time < 2 seconds
- **OE-6.3**: Routing & Optimization Module
  - Deployed as an internal microservice callable via HTTPS; expected compute time < 5 seconds per request under normal load
- **OE-6.4**: Email & Notification Services
  - SMTP over TLS (port 587) for transactional emails (account verification, password resets, scheduled reports)
  - WebSocket endpoint on port 443 for real-time in-app alerts (weather warnings, price spikes, data ingestion failures)

OE-7: Browser Interaction Requirements

- **OE-7.1**: File Downloads & Printing
  - PDF/CSV exports should invoke the native browser download dialog
  - Print dialogs for reports must format for A4 paper on Windows, macOS, and Linux

2.4 Design and Implementation Constraints

Machine-Learning & Data Manipulation: All data preprocessing, analytics, and machine-learning

models shall be implemented in **Python**.
Backend Services: Core API and business-logic services shall be written in Node.js/Python.
Frontend Framework: The user interface shall be built using React.js, with an evaluation of Next.js for server-side rendering or static export based on performance and SEO requirements.
Database Engine: The database platform is **not yet finalized**; candidate systems include PostgreSQL, MySQL, or MongoDB. The final choice must support horizontal scaling and standard backup/restore workflows.
Code Management: All source codes will be maintained in Git/GitHub.

3. Requirement Identifying Technique

This section describes the requirements identifying technique(s) which further help to derive functional requirements specification. The selection of the technique(s) will depend on the type of project. For instance,

3.1 Use Cases

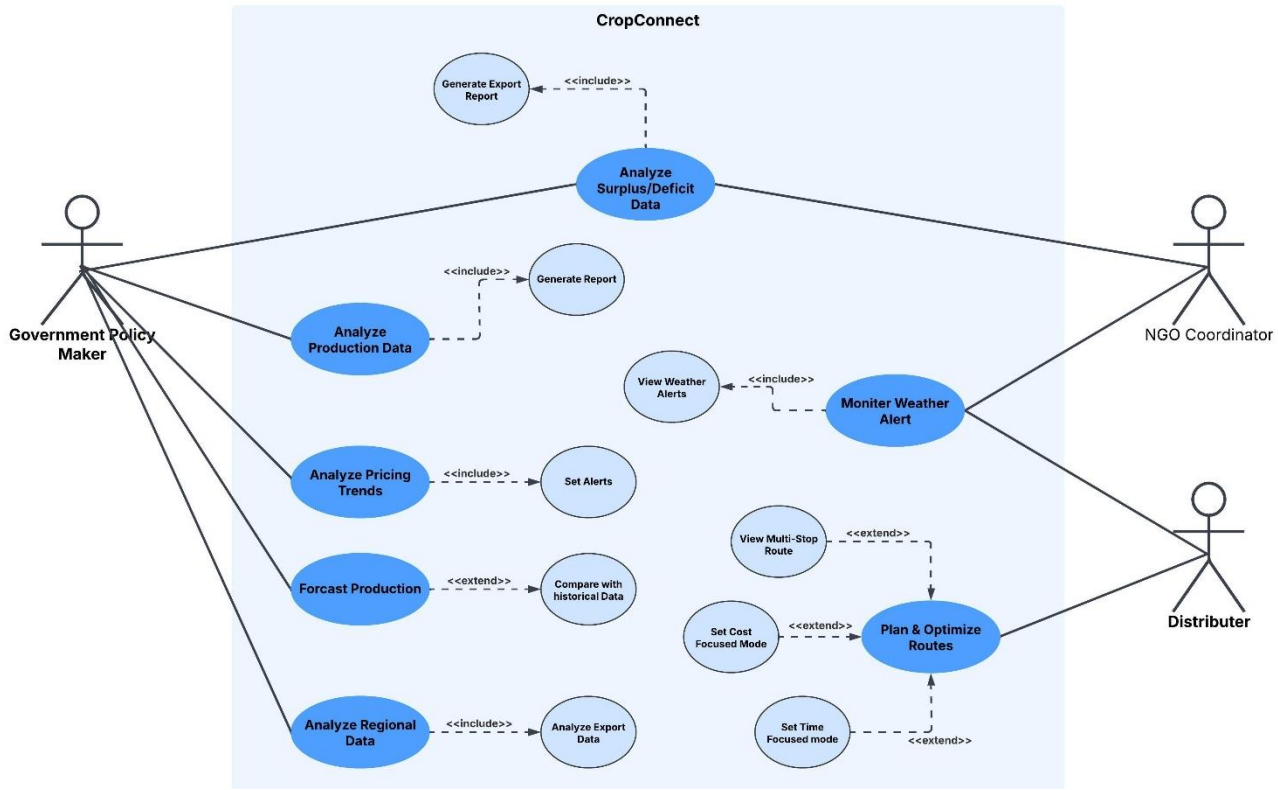


Fig A-2: Use Case Diagram of CropConnect System

3.1.1 Use Case 1: Analyze Production Data

Field	Details
Use Case ID	UC-1
Use Case Name	Analyze Production Data
Actors	Primary: Government Policy Maker
Description	A Policy Maker monitors district-wise crop production trends via tables and charts to inform agricultural policy.
Trigger	User navigates to the Production Data Analysis dashboard.
Preconditions	PRE-1: User authenticated as Policy Maker. PRE-2: District-level production data ingested.
Postconditions	POST-1: Exported report or set alerts for production changes.
Normal Flow	1.0 **Analyze Production Data** 1.1 The Government Policy Maker opens the Production Data Analysis Tab. 1.2 The system displays the Production Data Analysis Tab. 1.3 The system retrieves available regions from the database. 1.4 The user selects a region ("All" or a specific region). 1.5 The user filters data by crop type and time period. 1.6 The system filters production data based on the selected crop type and time period. 1.7 The system generates and displays a table and chart showing crop volumes, yield/ha, and trends. 1.8 The user drills into district details for more granular information. 1.9 The system retrieves and displays district-level data for the selected district. 1.10 The user compares current data with historical trends. 1.11 The system compares the selected data with historical trends and presents the comparison. 1.12 The user exports the report in PDF/CSV format. 1.13 The system generates and exports the report in the desired format. 1.14 The user sets alerts for production changes. 1.15 The system stores the user's alert settings and triggers notifications when production changes meet the defined criteria.
Alternative Flow	**2.0 No Data for Selected Filters** 2.1 At step 1.6, if the selected region/crop/time yields no records, the system displays "No data available for the chosen parameters." 2.2 The user adjusts filters or selects "All" to view broader data. **3.0 Retry After Data Load Failure** 3.1 At PRE-2, if data ingestion is incomplete, the system displays "Production data is not up to date. Retry loading?" 3.2 The user selects "Retry," and the system re-attempts data ingestion from the source.
Exceptions	**4.0 E1: Data Load Fail** 4.1 During step 1.6 (or when retrieving data), if the database query fails (e.g., timeout or missing table), the system displays "Error loading production data. Please retry."

Field	Details
	4.2a If the user clicks “Retry,” the system returns to step 1.6. 4.2b If the user cancels, the use case terminates. **5.0 E2: Export Error** 5.1 At step 1.12, if the PDF/CSV generator fails (e.g., disk full or format exception), the system shows “Export failed: [cause].” 5.2a If the user clicks “Try Again,” the system returns to step 1.12. 5.2b If the user cancels, the use case terminates without exporting.
Business Rules	Data must refresh whenever new data comes.
Assumptions	Data is accurate to 95%. Visualization renders within 3 seconds.

3.1.2 Use Case 2: Analyze Pricing Trends

Field	Details
Use Case ID	UC-2
Use Case Name	Analyze Pricing Trends
Actors	**Primary:** Government Policy Maker
Description	A Policy Maker examines historical and current price movements by crop and region to guide market interventions.
Trigger	User selects the Pricing Trends module.
Preconditions	PRE-1: User authenticated. PRE-2: Daily price data loaded.
Postconditions	POST-1: Price report exported or alerts set.
Normal Flow	1.0 **Analyze Pricing Trends** 1.1 The Government Policy Maker opens the Pricing Trends Analysis Tab. 1.2 The system opens the Pricing Trends Analysis module. 1.3 The system retrieves the list of crops and regions from the database. 1.4 The user selects the crop(s) and region(s) to analyze. 1.5 The user selects the time span for the analysis. 1.6 The system retrieves pricing data for the selected time span and filters it accordingly. 1.7 The system generates and displays trend charts showing price movements over time. 1.8 The user compares pricing trends across different regions.

Field	Details
	1.9 The system highlights discrepancies in pricing trends across regions. 1.10 The user identifies anomalies in the price trends. 1.11 The system identifies and highlights any anomalies or outliers in the pricing data. 1.12 The user exports the pricing report. 1.13 The system generates and exports the requested price report in PDF/CSV format.
Alternative Flows	**2.0 Incomplete Data** 2.1 At step 1.7, if some dates have missing price entries, the system overlays gaps in the chart and displays "Incomplete data for selected period." 2.2 The user may choose to interpolate missing dates or adjust the time span. **3.0 Overlay Multiple Crops** 3.1 After step 1.4, the user selects "Compare multiple crops." 3.2 The system overlays price trend lines of selected crops on the same chart. **4.0 Auto-Highlight Spikes/Dips** 4.1 At step 1.10, the system automatically detects price spikes/dips and highlights them in the chart with annotations. 4.2 The user clicks an annotation to view details (date, region, potential cause). **5.0 Configure Multi-Level Alerts** 5.1 Before step 1.12, the user clicks "Set Alerts" and defines threshold levels for price changes. 5.2 The system stores alert settings and will trigger notifications if price movements exceed the defined thresholds.
Exceptions	**6.0 E1: Data Fetch Fail** 6.1 During step 1.6, if price data cannot be retrieved (e.g., API timeout), the system displays "Unable to load price data. Please retry." 6.2a If the user clicks "Retry," the system returns to step 1.6. 6.2b If the user cancels, the use case terminates. **7.0 E2: Export Fail** 7.1 At step 1.12, if report generation fails (e.g., file permission error), the system shows "Export failed: [cause]." 7.2a If the user clicks "Try Again," the system returns to step 1.12. 7.2b If the user cancels, the use case terminates without exporting.
Business Rules	Alerts on any rate change.
Assumptions	Charts render within 3s. Data is Fetched and Displayed with in 1-2s

3.1.3 Use Case 3: Forecast Production

Field	Details
Use Case ID	UC-3
Use Case Name	Forecast Production
Actors	**Primary:** Government Policy Maker
Description	A Policy Maker generates AI-powered seasonal yield predictions using weather and historical data to support planning.
Trigger	User opens the Forecasting Tab.
Preconditions	PRE-1: Historical production & weather data available. PRE-2: Forecast models trained and deployed.
Postconditions	POST-1: Forecast results stored & exported.
Normal Flow	**1.0 Forecast Production** 1.1 The Government Policy Maker opens the Forecasting Tab/Page. 1.2 The system opens the Forecasting dashboard. 1.3 The system retrieves available seasons and forecasting horizons for selection. 1.4 The user selects the season and forecasting horizon. 1.5 The user selects the regions and crops for the forecast. 1.6 The system retrieves region and crop options from the database. 1.7 The user runs the AI model to generate the yield predictions. 1.8 The system runs the AI model using historical and weather data to generate yield predictions. 1.9 The system displays the predicted yields along with confidence intervals. 1.10 The user compares the forecasted yields with historical data. 1.11 The system compares the predicted yields with historical data and presents the comparison. 1.12 The user exports the forecast results in a report. 1.13 The system generates and exports the forecast results in CSV/PDF format.
Alternative Flows	**2.0 Insufficient Weather Data** 2.1 At step 1.8, if weather data for the selected season or region is incomplete, the system displays “Insufficient weather data. Using historical averages as fallback.” 2.2 The AI model runs using averaged weather inputs and flags results with a “Fallback” badge. **3.0 Adjust Weather Parameters & Rerun** 3.1 After step 1.9, if the user deems confidence intervals too wide, they click “Refine Model.” 3.2 The system prompts the user to adjust weather parameter weights or select alternative data sources. 3.3 The user modifies parameters and clicks “Rerun Forecast,” returning to step 1.7.
Exceptions	**4.0 E1: Model Error** 4.1 During step 1.8, if the AI model fails (e.g., runtime exception), the system shows “Forecasting error. Please try again.”

Field	Details
	4.2a If the user clicks “Retry,” the system returns to step 1.8. 4.2b If the user cancels, the use case terminates. **5.0 E2: Data Gap** 5.1 Before step 1.7, if historical production data is missing for one year or more, the system displays “Historical data gap detected. Forecast may be unreliable.” 5.2a The user may click “Proceed Anyway,” continuing to step 1.7. 5.2b The user may click “Cancel,” terminating the use case.
Business Rules	BR-1: Horizon ≤12 months.
Assumptions	A1: Model completes within 60s-90s. A2: Historical data covers at least 5 years.

3.1.4 Use Case 4: Analyze Regional Data

Field	Details
Use Case ID	UC-5
Use Case Name	Analyze Regional Data
Actors	**Primary:** Government Policy Maker
Description	A Policy Maker compares key metrics (production, price) across multiple regions to guide resource allocation.
Trigger	User opens the Regional Analysis dashboard.
Preconditions	PRE-1: Multi-regional data synchronized. PRE-2: Comparison metrics defined.
Postconditions	POST-1: Exported comparison report; benchmarks set.
Normal Flow	**1.0 Analyze Regional Data** 1.1 The Government Policy Maker opens the Regional Analysis dashboard. 1.2 The system opens the Regional Analysis dashboard. 1.3 The system retrieves available regions and metrics from the database. 1.4 The user selects regions and metrics for comparison. 1.5 The user selects the time period for comparison (e.g., last year, last quarter). 1.6 The system retrieves data for the selected time period and prepares the comparison. 1.7 The system generates and displays side-by-side charts and tables comparing the selected regions’ performance. 1.8 The user identifies performance gaps across regions. 1.9 The system highlights performance gaps and areas of concern. 1.10 The user generates and exports a comparison report. 1.11 The system generates and exports the report. 1.12 The user saves the monitoring targets for future comparison. 1.13 The system saves the monitoring targets for future reference.

Field	Details
Alternative Flows	**2.0 Data Mismatch** 2.1 At step 1.6, if the retrieved data for any region does not align (e.g., inconsistent units or missing dates), the system displays “Data mismatch detected.” 2.2 The user may choose “Proceed with available data” (flagged) or “Cancel” to adjust filters. **3.0 Export Retry** 3.1 At step 1.10, if the report export process fails, the system automatically retries up to two times. 3.2 If all attempts fail, the system displays “Export failed. Please try again later.”
Exceptions	**4.0 E1: Data Mismatch** 4.1 During step 1.6, if the system detects data from two regions with conflicting measurement units (e.g., metric vs. imperial), it shows “Cannot compare due to inconsistent units.” 4.2a The user clicks “Resolve Units,” returning to step 1.4 to reselect or convert units. 4.2b If the user cancels, the use case terminates. **5.0 E2: Export Fail** 5.1 At step 1.10, if the file generator throws an exception (e.g., write permission denied), the system displays “Export failed: [cause].” 5.2a The system retries automatically. 5.2b If retries are exhausted, the user can click “Try Again” (returning to step 1.10) or “Cancel,” terminating the use case.
Business Rules	BR-1: Must select ≥2 regions. BR-2: Data >30 days old flagged.
Assumptions	A1: Data latency ≤24h. A2: Rendering time ≤10s.

3.1.5 Use Case 5: Plan and Optimize Routes

Field	Details
Use Case ID	UC-6
Use Case Name	**Plan and Optimize Routes**
Actors	**Primary:** Distributor **Secondary**: NGO
Description	A Distributor (and **NGO** when they have to supply) computes the most efficient transport route between source and destination regions, balancing cost, distance, and weather.

Field	Details
Trigger	User clicks **Calculate Route** after selecting source/destination in the Transport Routes module.
Preconditions	PRE-1: Distributor authenticated.PRE-2: Latest fuel, toll, road, and weather data available.PRE-3: Optimization engine operational.
Postconditions	POST-1: Route displayed on map and saved.POST-2: ETA and cost logged.
Normal Flow	**1.0 Plan and Optimize Routes**1.1 The Distributor (or NGO) selects the source and destination regions. 1.2 The user clicks “Calculate Route” to start the route optimization process. 1.3 The system gathers fuel, toll, road, and weather data. 1.4 The system calculates the best transport route based on cost, distance, and weather. 1.5 The system displays the optimized route, including directions, ETA, and cost. 1.6 The user reviews and adjusts the route (e.g., add stops or change focus to cost/time). 1.7 The user saves the optimized route for future reference. 1.8 The system saves the optimized route and logs ETA and cost.
Alternative Flows	**2.0 Multi-Stop** 2.1 At step 1.2, before clicking “Calculate Route,” the user clicks “Add Stops” and inputs one or more intermediate waypoints. 2.2 The system recalculates the optimized path including all stops and returns to step 1.4. **3.0 Cost-Focused** 3.1 At step 1.2, the user toggles “Cost-Focused” mode. 3.2 The system prioritizes minimum fuel and toll costs over shortest distance when calculating the route and proceeds to step 1.4. **4.0 Time-Focused** 4.1 At step 1.2, the user toggles “Time-Focused” mode. 4.2 The system prioritizes shortest ETA (even if cost is higher) and proceeds to step 1.4.
Exceptions	**5.0 E1: No Viable Route** 5.1 During step 1.4, if the optimization engine finds no feasible path (e.g., disconnected roads), the system displays “No viable route available.” 5.2 The user may select alternate regions or click “Cancel,” terminating the use case. **6.0 E2: Severe Weather** 6.1 During step 1.3, if any segment has a “Severe” weather alert, the system warns “Severe weather detected. Reroute recommended.” 6.2a If the user clicks “Reroute,” the use case returns to step 1.2 (with option to modify constraints).

Field	Details
	6.2b If the user clicks “Proceed Anyway,” the system continues to step 1.4 but logs a severe-weather warning.
Business Rules	BR-1: Cost = distance×fuel_price + tolls. BR-2: Flag severe-weather segments.
Assumptions	A1: Road data updated weekly. A2: Calculation completes ≤15s.

3.1.6 Use Case 6: Monitor Weather Alerts

Field	Details
Use Case ID	UC-7
Use Case Name	Monitor Weather Alerts
Actors	**Primary:** Distributor **Secondary**: NGOs
Description	A Distributor & NGO views real-time weather warnings along planned routes and can adjust plans accordingly.
Trigger	User opens the Weather Alerts dashboard or receives a push notification.
Preconditions	PRE-1: Distributor authenticated.PRE-2: Weather alerts feed active.PRE-3: Route data loaded.
Postconditions	POST-1: Weather-related adjustments logged.
Normal Flow	**1.0 Monitor Weather Alerts** 1.1 The Distributor (or NGO) opens the Weather Alerts dashboard. 1.2 The system opens the Weather Alerts dashboard. 1.3 The system retrieves weather alerts for the user’s route and displays them. 1.4 The user views weather warnings for the planned routes. 1.5 The user decides whether to proceed, delay, or reroute based on the displayed weather data. 1.6 If rerouting is needed, the user invokes Route Planning & Optimization. 1.7 The system logs the weather-related adjustment and notifies relevant parties. 1.8 The user communicates changes to the driver (if needed).
Alternative Flows	**2.0 Auto-Suggest Reroute on Severe Alert** 2.1 At step 1.5, when the system identifies a “Severe” weather alert, it automatically computes an alternate route. 2.2 The user can accept or reject the suggested reroute. **3.0 Suggest Emergency Shelter if No Safe Route Found** 3.1 If no viable reroute exists due to widespread severe weather, the system displays “No safe route found. Show nearby emergency shelters?” 3.2 The user selects “Yes,” and the system lists the nearest shelters.

Field	Details
Exceptions	**E1: Alert Feed Unavailable** 1. At step 2, if the weather feed cannot be retrieved (e.g., service down), the system displays “Weather alerts currently unavailable. Last known data shown.” 2a. The user may close the dashboard or try again later.
Business Rules	BR-1: Only “Severe” and higher-level alerts are shown by default.
Assumptions	A1: Alerts refresh every 15m.

3.1.7 Use Case 7: Analyze Surplus and Deficit

Field	Details
Use Case ID	UC-SDA-013
Use Case Name	Analyze Surplus and Deficit
Actors	**Primary:** NGO Coordinator
Description	An NGO Coordinator assesses regional food-security imbalances by comparing production vs. consumption.
Trigger	User opens Surplus/Deficit Analysis.
Preconditions	PRE-1: Production & consumption data ingested.
Postconditions	POST-1: Critical regions flagged; report exported.
Normal Flow	**1.0 Analyze Surplus and Deficit** 1.1 The NGO Coordinator opens the Surplus/Deficit Analysis dashboard. 1.2 The system opens the Surplus/Deficit Analysis dashboard. 1.3 The system retrieves and displays available regions, crops, and time periods for selection. 1.4 The user selects regions, crops, and time periods for comparison. 1.5 The user computes surplus/deficit data for the selected regions. 1.6 The system computes surplus/deficit data based on production and consumption. 1.7 The system displays surplus/deficit results on a map and in a table format. 1.8 The user identifies critical areas that need intervention. 1.9 The user exports the report. 1.10 The system generates and exports the report in the requested format.
Alternative Flows	**2.0 Consumption Data Missing** 2.1 At step 1.5, if consumption data for a region or time period is missing, the system displays “Consumption data unavailable. Using estimate based on previous months.” 2.2 The system computes surplus/deficit using estimated values and flags those entries with an “Estimate” badge.

Field	Details
Exceptions	**3.0 E1: Consumption Data Missing** 3.1 During step 1.5, if no historical consumption data exists, the system warns "No consumption data. Cannot compute deficit reliably." 3.2 The user may click "Use Estimates" (proceeding to step 1.6) or "Cancel," terminating the use case.
Business Rules	BR-1: A region is flagged as "Deficit" if consumption ≥ 120% of production.
Assumptions	A1: Consumption data monthly.

Event Response Table

Event	System State	Response
New fuel- or toll-rate feed arrives with any change	• Stored cost parameters reflect previous rates. • Active saved routes use outdated cost calculations.	1. Update cost-parameter store with the new rates. 2. Recalculate total cost for each saved route. 3. Push in-app notification to Distributors: "Fuel/Toll rates updated—recalculate routes for current costs."
Weather API reports a "Severe" or higher warning for a region	• Active routes may traverse the affected region. • No existing weather-hazard flags on those routes.	1. Mark the affected region as **Weather Hazard** in the routing database. 2. Recalculate alternative paths for any active or saved route crossing that region. 3. Push in-app notification: "Severe weather in Region X—view updated route."
Three consecutive failures ingesting an external data feed	• Ingestion failure count for that feed ≥ 3. • Latest data remains stale or incomplete.	1. Retry the ingestion job immediately. 2. If retry also fails, log the failure.

4.0 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes. For further detail see Chapter 10 "Documenting the requirements". Let consider the feature scheme as an example.

4.1 Real-Time Crop Data Collection

Identifier	FR-1.1
Title	Real-Time Crop Data Collection
Requirement	The system shall collect real-time crop data, including production volume, crop types, and location. The system shall update crop data automatically as new data is ingested.
Source	Government Policy Maker
Rationale	To ensure that Government Policy Makers have access to up-to-date crop production data for decision-making.
Business Rule (if required)	None
Dependencies	None
Priority	High

4.2 Historical & Real-Time Weather Data Integration

Identifier	FR-1.2
Title	Historical & Real-Time Weather Data Integration
Requirement	The system shall integrate historical and real-time weather data to evaluate the impact of weather on crop production. The system shall display weather trends alongside crop production data for comparative analysis.
Source	Government Policy Maker
Rationale	To support accurate forecasting and crop production analysis based on weather patterns.
Business Rule (if required)	None
Dependencies	None
Priority	High

4.3 Demand-Supply Trend Analysis

Identifier	**FR-1.3**
Title	Demand–Supply Trend Analysis
Requirement	The system shall analyze demand-supply trends for crops, highlighting any discrepancies between regions. The system shall generate reports for Government Policy Makers based on demand-supply gaps.
Source	Government Policy Maker
Rationale	To help Government Policy Makers identify regions that require intervention.
Business Rule (if required)	None
Dependencies	None
Priority	Medium

4.4 Seasonal Yield Prediction

Identifier	**FR-2.1**
Title	Seasonal Yield Prediction
Requirement	The system shall use machine learning models to predict seasonal crop yields based on historical and weather data. The system shall provide confidence intervals for each prediction.
Source	Government Policy Maker
Rationale	To provide accurate yield predictions for Government Policy Makers to plan future agricultural needs.
Business Rule (if required)	None
Dependencies	None
Priority	High

4.5 Forecast Comparison

Identifier	**FR-2.2**
Title	Forecast Comparison
Requirement	The system shall allow comparison between predicted crop yields and actual historical production. Users should be able to view discrepancies in visual formats such as graphs and charts.
Source	Government Policy Maker

Identifier	**FR-2.2**
Rationale	To help Government Policy Makers assess the accuracy of forecasts and improve future models.
Business Rule (if required)	None
Dependencies	None
Priority	Medium

4.6 Pricing Tends Analysis

Identifier	**FR-3.1**
Title	Pricing Trends Analysis
Requirement	The system shall allow users to analyze historical and current price movements for crops. The system shall enable comparison across multiple regions and time spans.
Source	Government Policy Maker
Rationale	To assist Government Policy Makers in making informed decisions about market interventions.
Business Rule (if required)	None
Dependencies	None
Priority	High

4.7 Price & Toll Alerts

Identifier	**FR-3.2**
Title	Price & Toll Alerts
Requirement	The system shall send real-time alerts to Distributors when there are significant changes in crop prices or transportation tolls. The system shall support the configuration of multi-level alerts for different thresholds.
Source	Distributor
Rationale	To allow Distributors to take quick action in response to market fluctuations and transportation costs.
Business Rule (if required)	None
Dependencies	None
Priority	Medium

4.8 Surplus/Deficit Region Identification

Identifier	**FR-4.1**
Title	Surplus/Deficit Region Identification
Requirement	The system shall identify surplus and deficit regions based on crop production and consumption data. The system shall display surplus/deficit data visually on a map and in tables.
Source	NGO Coordinator
Rationale	To help NGO Coordinators and Government Policy Makers allocate resources and plan interventions.
Business Rule (if required)	None
Dependencies	None
Priority	High

4.9 Regional Comparison

Identifier	**FR-4.2**
Title	Regional Comparison
Requirement	The system shall allow users to compare key metrics (production, price) across different regions. The system shall provide reports comparing multiple regions' agricultural data.
Source	Government Policy Maker
Rationale	To help Government Policy Makers allocate resources effectively across regions.
Business Rule (if required)	None
Dependencies	None
Priority	Medium

4.10 Transportation Route Planning

Identifier	**FR-4.3**
Title	Transportation Route Planning
Requirement	The system shall allow Distributors to plan the most efficient transportation routes between regions. The system shall provide optimizations based on cost, distance, and real-time weather data.

Identifier	**FR-4.3**
Source	Distributor
Rationale	To ensure Distributors minimize logistical costs and transportation time.
Business Rule (if required)	None
Dependencies	None
Priority	High

4.11 Interactive Dashboard Visualization

Identifier	**FR-5.1**
Title	Interactive Dashboard Visualization
Requirement	The system shall display production, pricing, forecast, comparison, and NGO relief data on an interactive dashboard. The dashboard shall allow users to filter and drill down data by region, crop type, and time period.
Source	Government Policy Maker, NGO, Distributor
Rationale	To allow users to make quick, informed decisions based on visualized data.
Business Rule (if required)	None
Dependencies	None
Priority	High

4.12 Export Reports

Identifier	**FR-5.2**
Title	Export Reports
Requirement	The system shall allow users to export reports in PDF or CSV formats for further analysis. The system shall support exporting dashboard snapshots to share with stakeholders.
Source	Government Policy Maker
Rationale	To provide stakeholders with reports that can be used for meetings, planning, and analysis.
Business Rule (if required)	None
Dependencies	None
Priority	Medium

4.13 Scheduled Email Reports

Identifier	**FR-5.3**
Title	Scheduled Email Reports
Requirement	The system shall allow users to schedule reports to be automatically emailed to stakeholders. Reports shall include up-to-date production data, market trends, and forecasts.
Source	Government Policy Maker
Rationale	To automate the distribution of reports to key stakeholders without manual intervention.
Business Rule (if required)	None
Dependencies	None
Priority	Medium

5.0 Non-Functional Requirements

5.1. Reliability

- **REL-1:** The system shall achieve a Mean Time Between Failures (MTBF) of at least **1,000 hours**.
- **REL-2:** In the event of a failure, the system shall automatically recover to a functional state within **2-3 minutes** and send an alert notification to administrators within **1 minute** of detecting the failure.
- **REL-3:** The system shall detect and log data-processing or service-availability errors within **1 minute** of occurrence, achieving a detection rate of **100%** for all critical failures..
- **REL-5:** The system shall refresh and ingest new data from external sources (PBS, MSN Weather) automatically at least every **24 hours**, with a maximum delay of **1 hour** from data availability to ingestion.

5.2. Usability

- **USE-1: 95%** of new users shall be able to complete onboarding (create an account, select role, configure initial settings) without assistance within **3 minutes**.
- **USE-2:** When an error occurs (e.g., failed data load, invalid input), the system shall display a clear error message and recovery instructions within **2 seconds**, guiding users to resolution steps.
- **USE-3:** Every form field and major interface control shall include contextual help or a tooltip accessible within **one click** or hover, covering **100%** of key functionality (e.g., report generation, filtering).
- **USE-4: 100%** of interactive elements (buttons, links, filters) shall be operable via keyboard only (Tab navigation, Enter/Space activation).

5.3. Performance

- **PER-1: 95%** of page loads (dashboard, report screens, map views) shall complete within **2 seconds** under normal operating conditions.
- **PER-2: 95%** of backend data-processing tasks (crop data analysis, price trend calculations, weather-data integration) shall finish within **30 seconds**, even under peak load.
- **PER-3:** The system shall sustain **500** concurrent authenticated users performing typical operations (viewing dashboards, generating reports) with average response time below **2 seconds**.
- **PER-4: 95%** of database queries for frequently accessed data (e.g., production trends, price histories) shall return results within **2 seconds**.

5.4. Security

- **SEC-1: 100%** of data exchanges between clients and servers shall use TLS 1.2 or higher, ensuring encryption in transit.
- **SEC-2:** The system shall enforce role-based access control (RBAC) so that **100%** of sensitive data endpoints verify user roles before granting access.
- **SEC-3:** User authentication shall be managed via JWTs with access tokens that expire after **15 minutes** and refresh tokens that expire after **7 days**, securing user sessions.

5.5. Maintainability

- **MNT-1:** The codebase shall follow a modular architecture: each module (Data Ingestion, Weather, Forecasting, Routing, Alerting, Reporting, Authentication) shall compile and deploy independently, with **≥90%** separation of concerns.
- **MNT-2: 100%** of application errors and at least **80%** of key user actions shall be logged (timestamp, user ID, context) to facilitate debugging and troubleshooting.
- **MNT-3:** Public functions and modules shall include inline documentation covering **≥90%** of method signatures, parameters, and expected behavior.
- **MNT-4:** The entire codebase shall reside in a Git repository; **100%** of changes must be committed, code-reviewed, and merged via pull requests.

5.6. Scalability

- **SCL-1:** The application shall support horizontal scaling by adding additional application servers within **30 minutes**, doubling user-concurrency capacity when needed.
- **SCL-2:** The (PostgreSQL) database shall support horizontal scaling (read replicas) to maintain **<2 seconds** query response times for datasets up to **1 TB**.
- **SCL-3:** During seasonal traffic spikes (e.g., harvest season with 2× normal load), the system shall maintain average response times under **3 seconds** for dashboards and report generation.

5.7. Compatibility

- **CMP-1:** The web application shall be fully functional on the two latest major versions of Chrome, Firefox, and Edge; all core features must pass **100%** of compatibility tests on each browser.
- **CMP-2:** The system shall successfully integrate with external APIs (MSN Weather, PBS ingestion) with a **≥99%** call success rate over any rolling 30-day period.

5.8. Availability

- **AV-1:** In the event of a primary server failure, failover to a standby server shall complete within **5 minutes** to minimize service interruption.
- **AV-2:** The system's annual availability shall be at least **99.9%**, excluding scheduled maintenance.

6.0 External Interface Requirements

This section provides information to ensure that the system will communicate properly with users and with external hardware or software elements. A complex system with multiple subcomponents should create a separate interface specification or system architecture specification. The interface documentation could incorporate material from other documents by reference. For instance, it could point to a hardware device manual that lists the error codes that the device could send to the software.

6.1 User Interfaces Requirements

6.1.1 General Styling and Branding Rules

- **Design System**
  Follow the CropConnect Agricultural UI Design System v1.0.
- **Brand Colors**
  - **Primary**: Teal #0d9488 – used for main actions and primary navigation elements.
  - **Secondary**: Green #22c55e – indicates success states and positive confirmations.
  - **Accent**: Orange #f97316 – reserved for warnings or elements that require extra attention.
  - **Error**: Red #ef4444 – used for error messages and critical alerts.
  - **Background**: Light gray #f9fafb – background color for main content areas.
- **Typography**
  - **Primary font Family**: Inter, system-ui, sans-serif
  - **Body text**: 14 px, regular weight
  - **Headings**:
    - H1 – 24 px, bold
    - H2 – 20 px, semibold
    - H3 – 18 px, semibold
  - **Small text**: 12 px for captions, metadata, and helper text
- **Icon Set**
  Use Lucide React icon library for all interface icons to ensure consistency.

6.1.2 Layout and Resolution Constraints

- **Desktop Web**
  - Minimum resolution: 1280 × 720 px

6.1.3 Navigation and Standard Controls

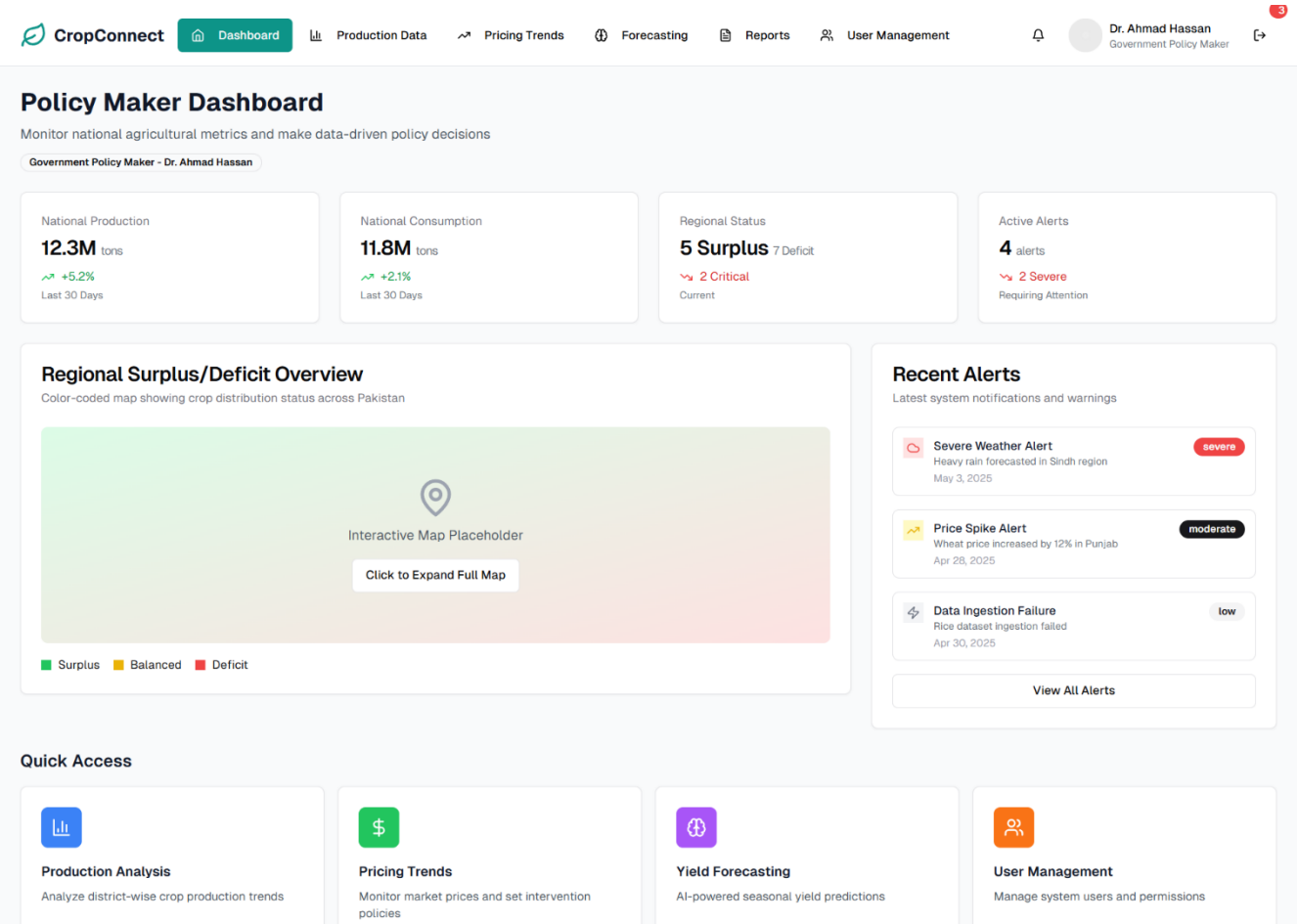
- **Global Header**
  Every page (including dashboards and detail views) must include:
  1. CropConnect logo on the left (clickable, returns to home/dashboard)
  2. Main navigation menu in the center (aligned horizontally)
  3. User menu on the right, containing:
     - User's name and role badge
     - Notifications bell icon
     - Profile dropdown (with Settings, Help, Logout)
- **User Menu**
  - Displays the logged-in user's name and role badge.
  - Dropdown options:
    - **Settings** (personal preferences, account details)
    - **Help Documentation** (opens contextual help)
    - **Logout** (ends the session)
- **Message Formatting**
  - **Success messages**:
    - Green banner with checkmark icon
    - Auto-dismiss after 5 seconds
  - **Error messages**:
    - Red banner with exclamation-mark icon
    - Manual dismiss required (click "X" or "Dismiss")
  - **Warning messages**:
    - Orange banner with warning-triangle icon
    - Auto-dismiss after 8 seconds
  - **Info messages**:
    - Blue banner with info icon
    - Auto-dismiss after 6 seconds

6.1.4 Role-Based Interface Variations

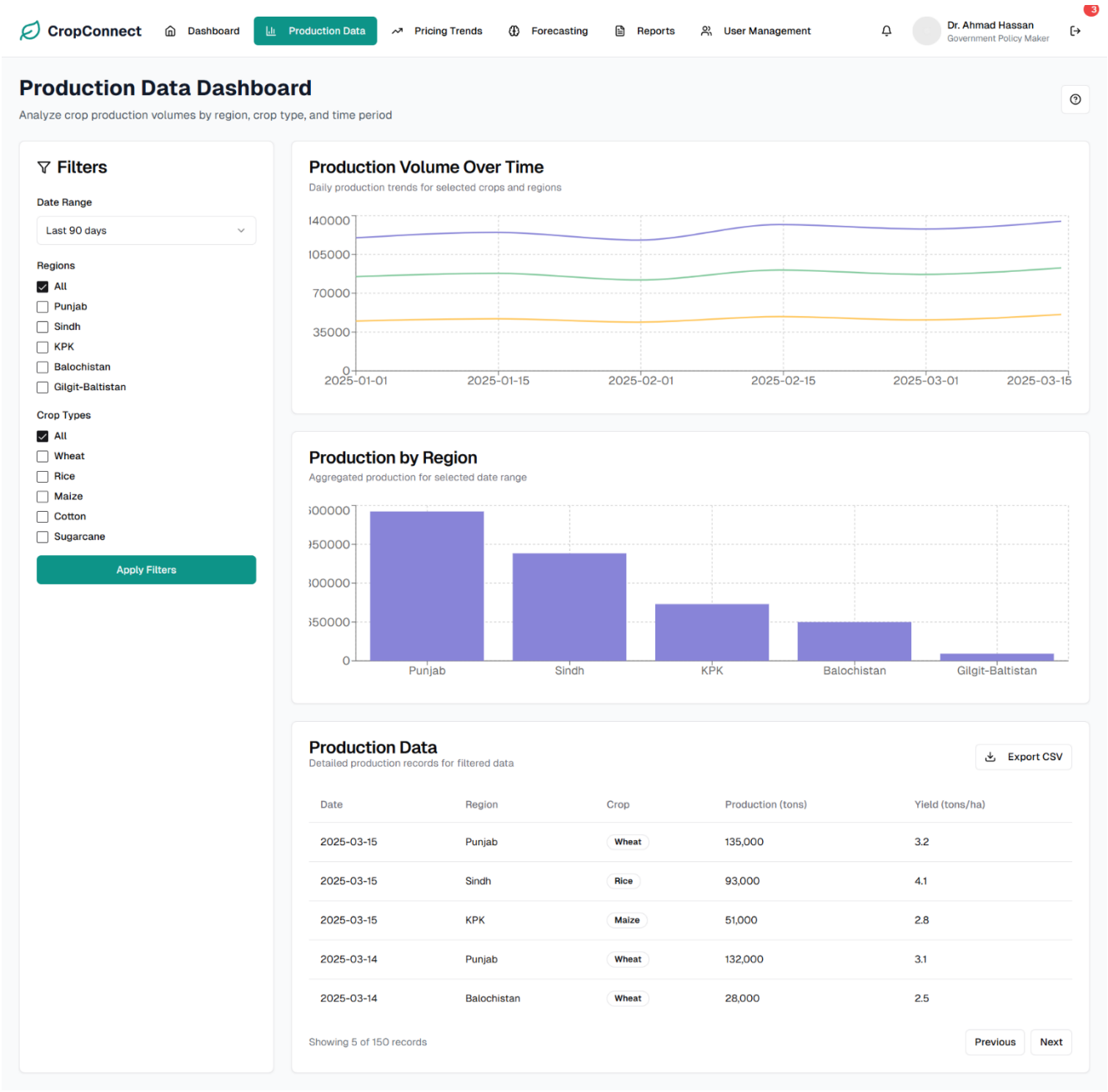
1. **Government Policy Maker**
   - Has full access to all modules, including:
     - User management
     - System administration
     - Production Data Analysis
     - Pricing Trends Analysis
     - Production Forecasting
     - Regional Analysis

        - All reporting/alert configuration
2. **NGO Coordinator**
    - Access limited to:
        - Surplus/Deficit Analysis
        - Route Planning & Optimization
        - Weather Alert Monitoring
        - Reporting and exports for NGO-related dashboards
3. **Distributor**
    - Access limited to:
        - Route Planning & Optimization
        - Weather Alert Monitoring
        - Pricing Trends Analysis

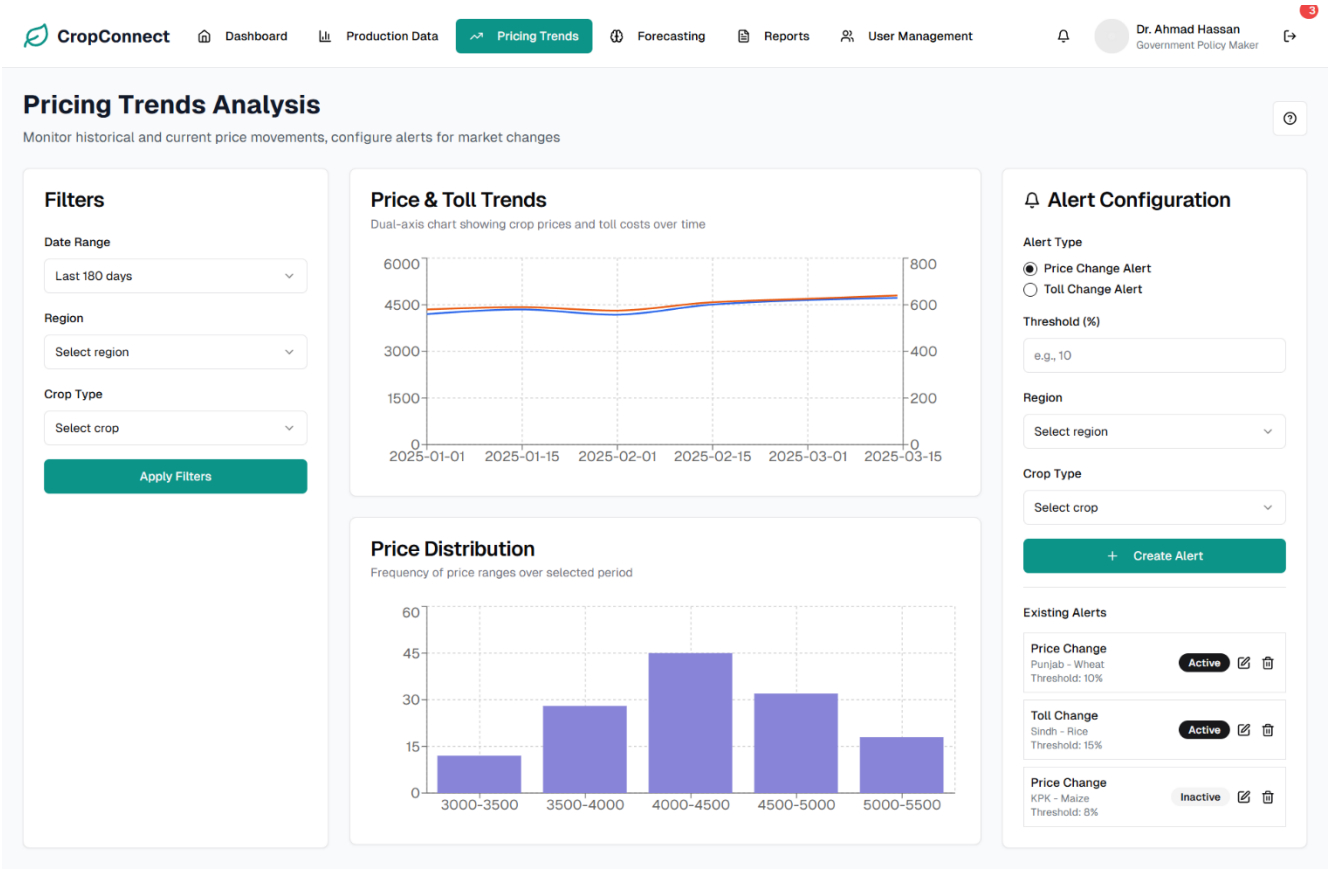
6.1.5 Mockups



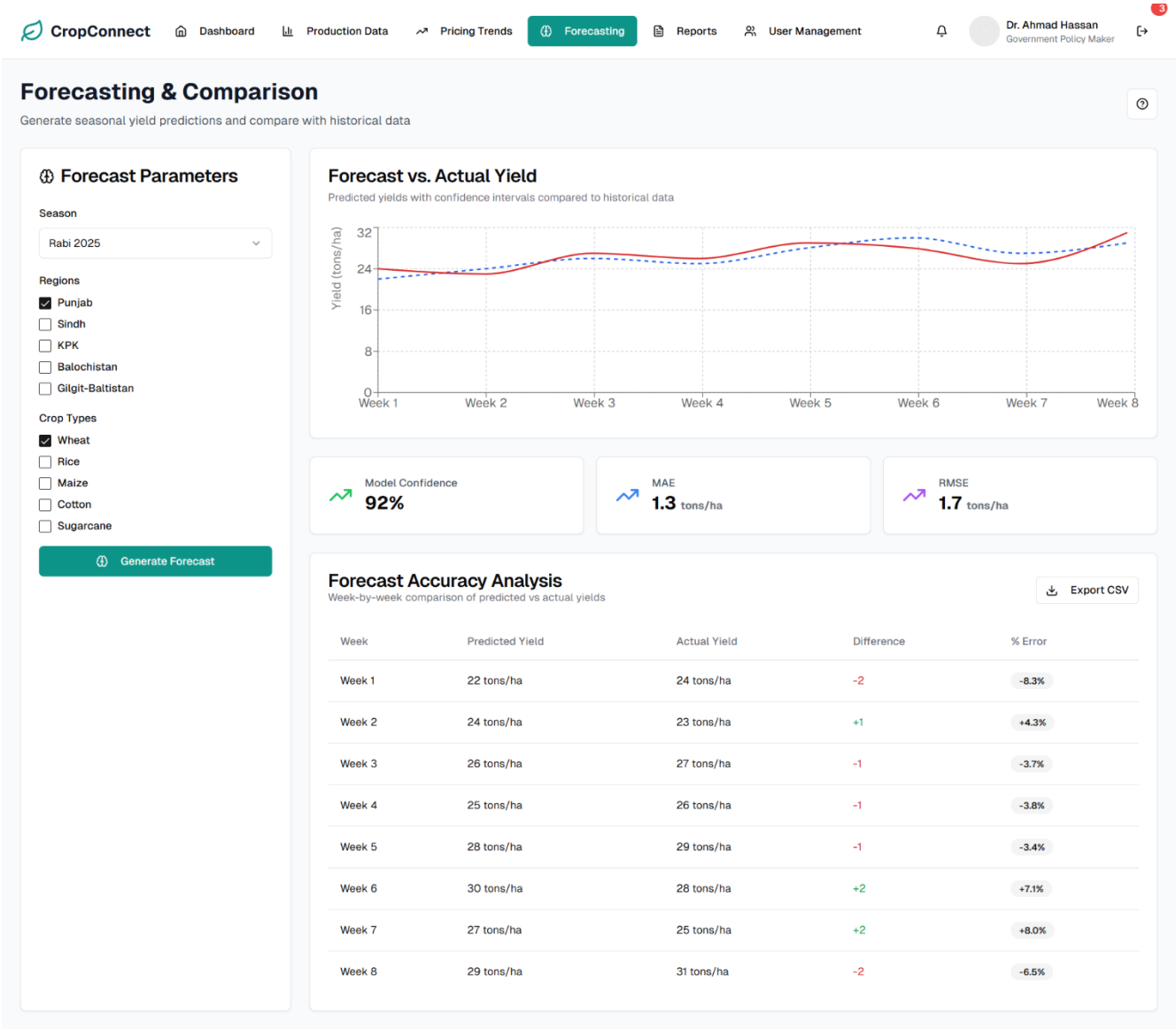
Dashboard of Government Policy Maker



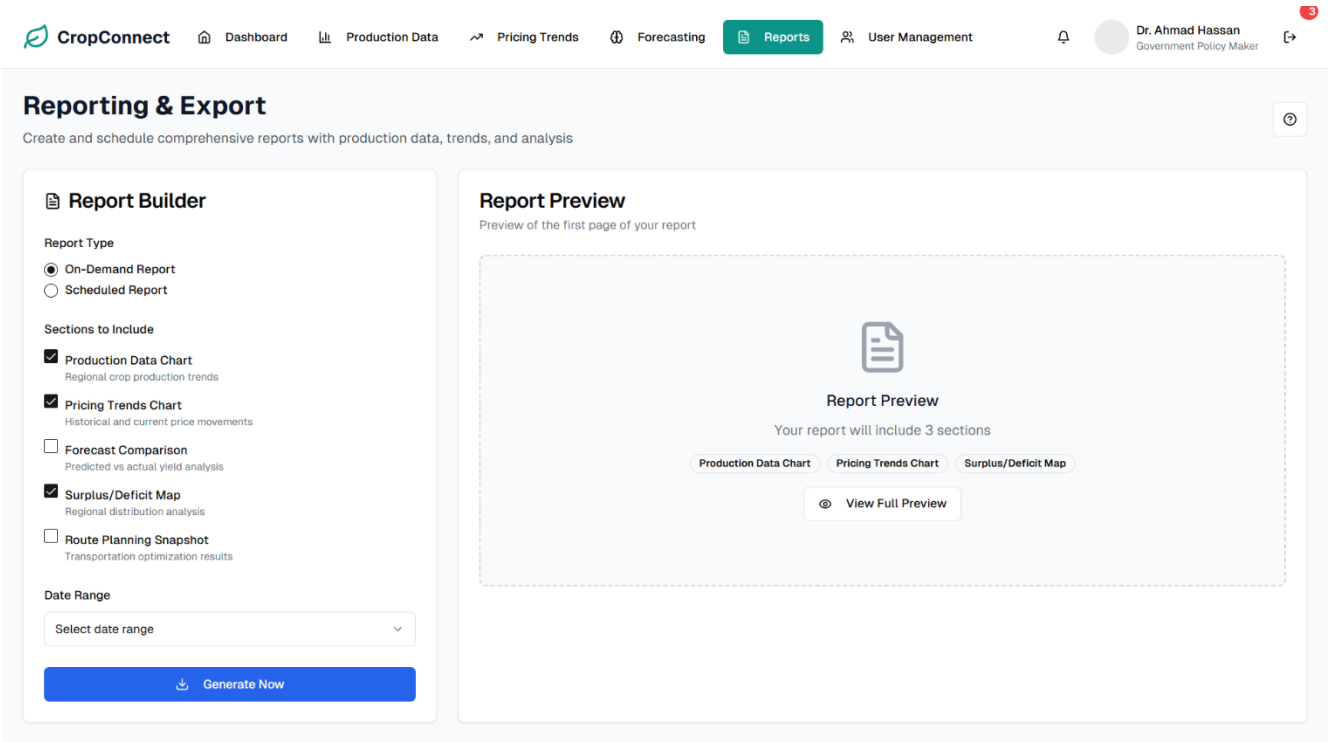
Production Data Interface of Government Policy Maker



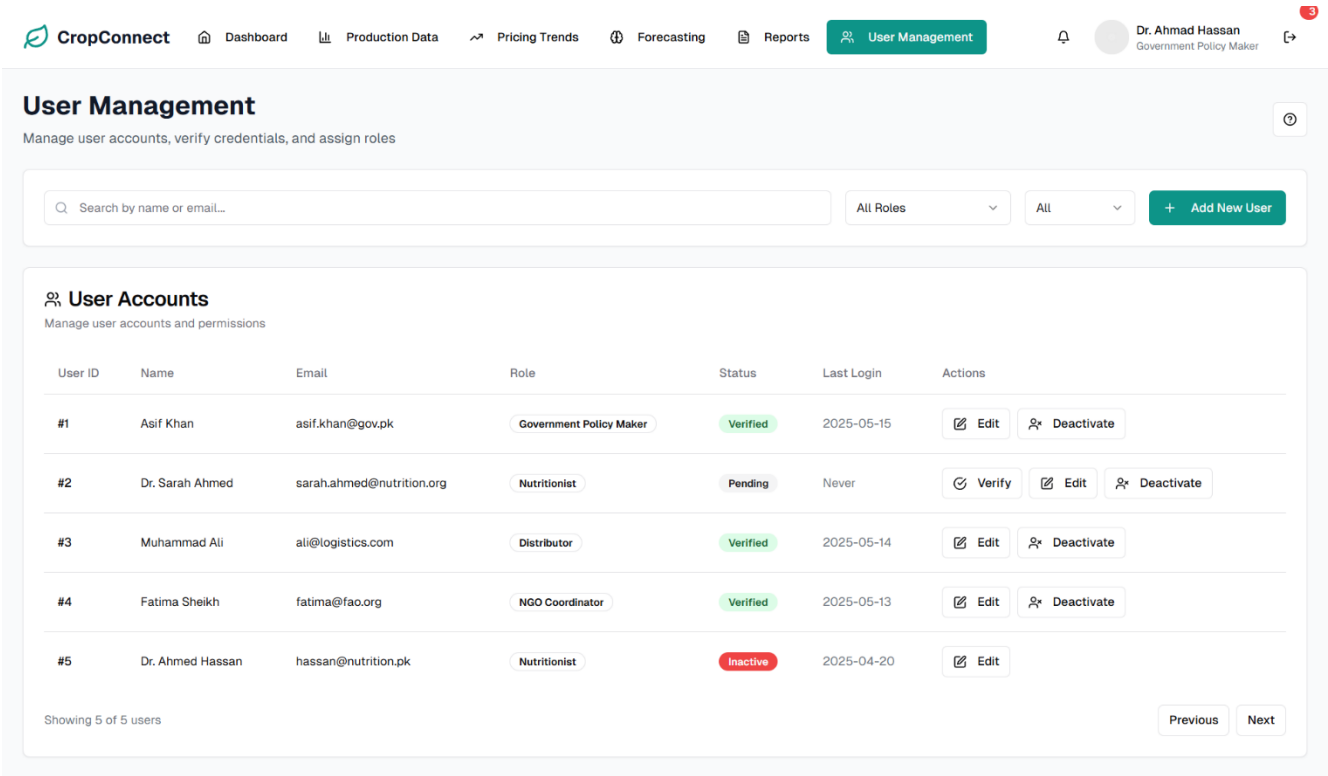
Pricing Trend Analysis Interface of Government Policy Maker



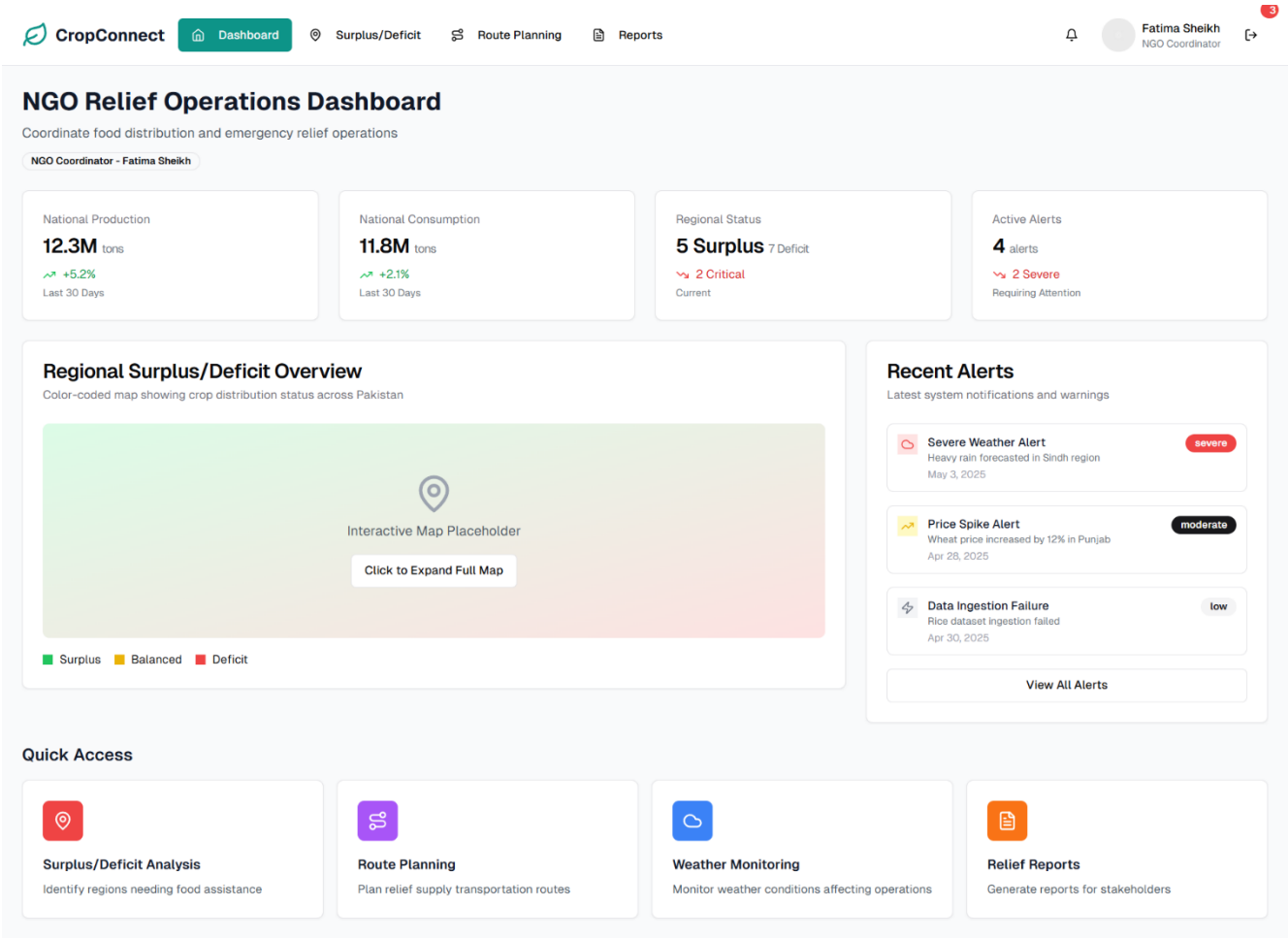
Forecasting & Comparison Data Interface for Government Policy Maker



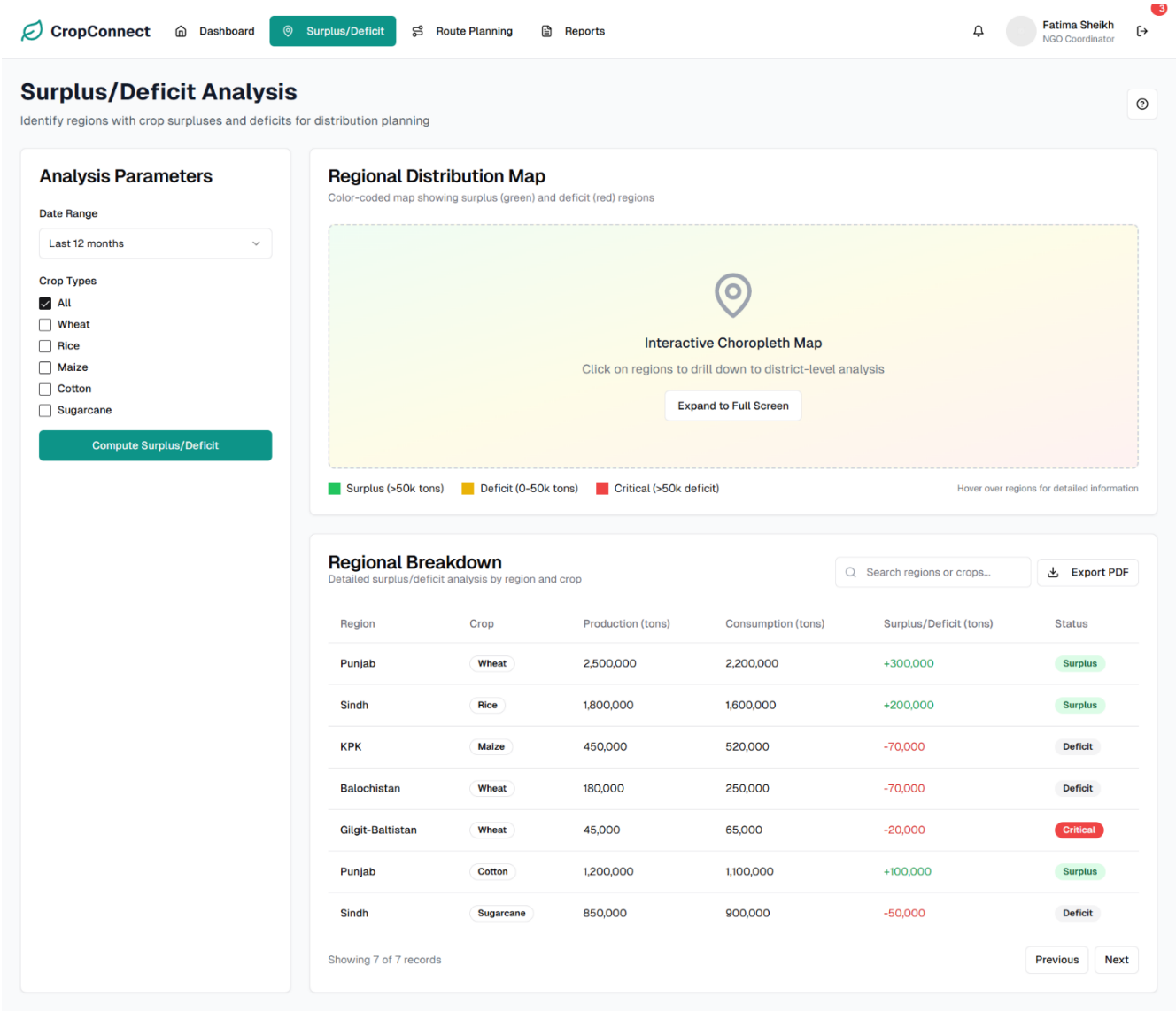
Reports



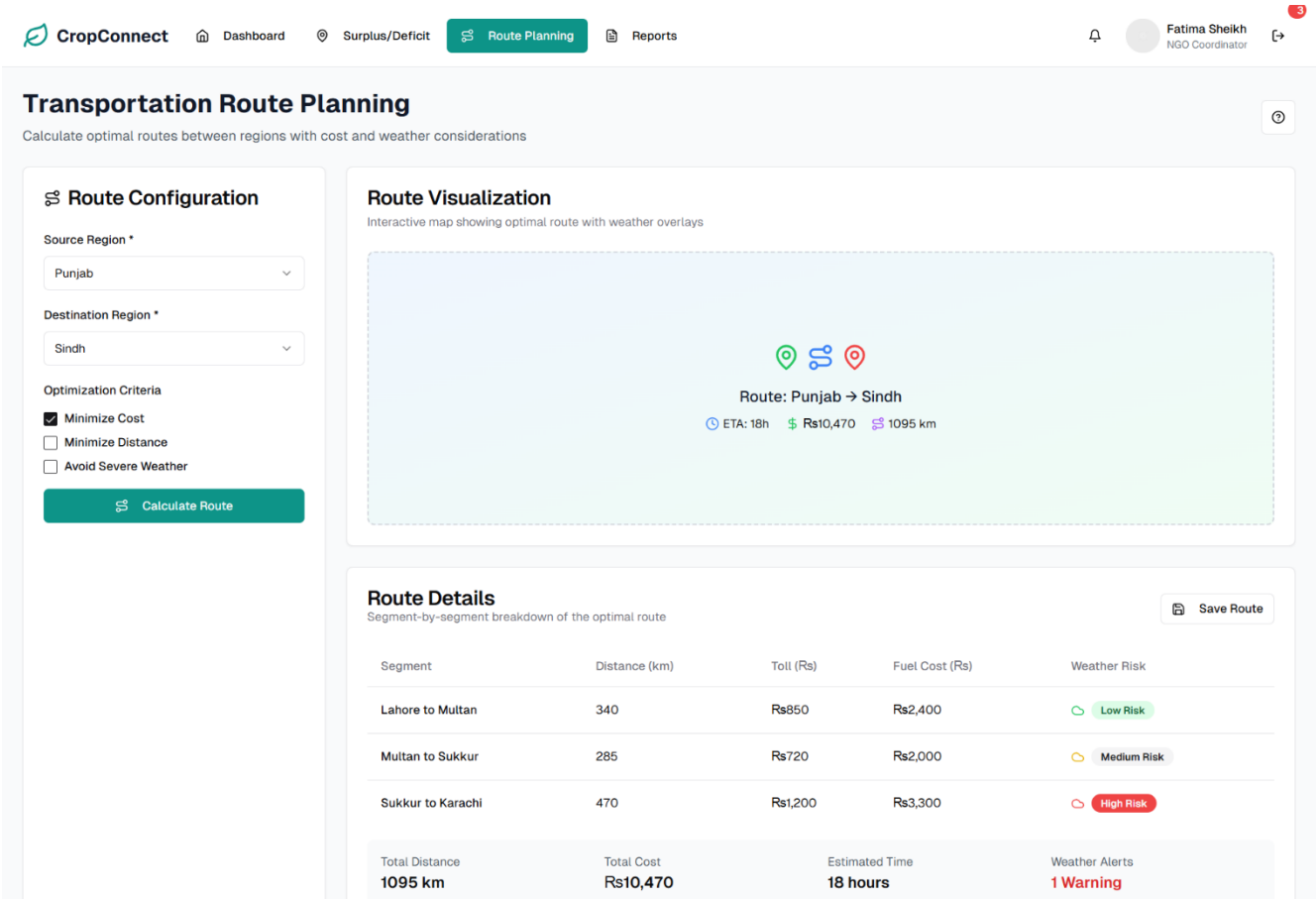
User Management

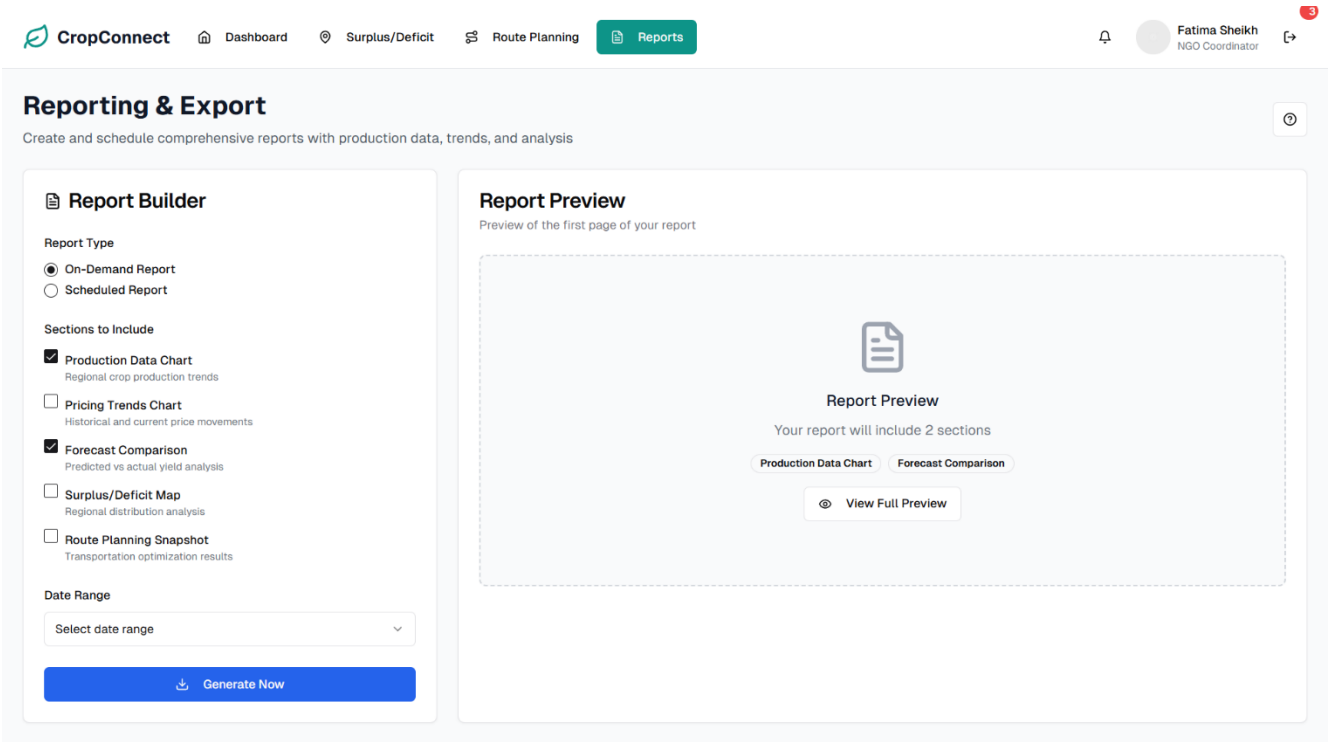


NGO Dashboard

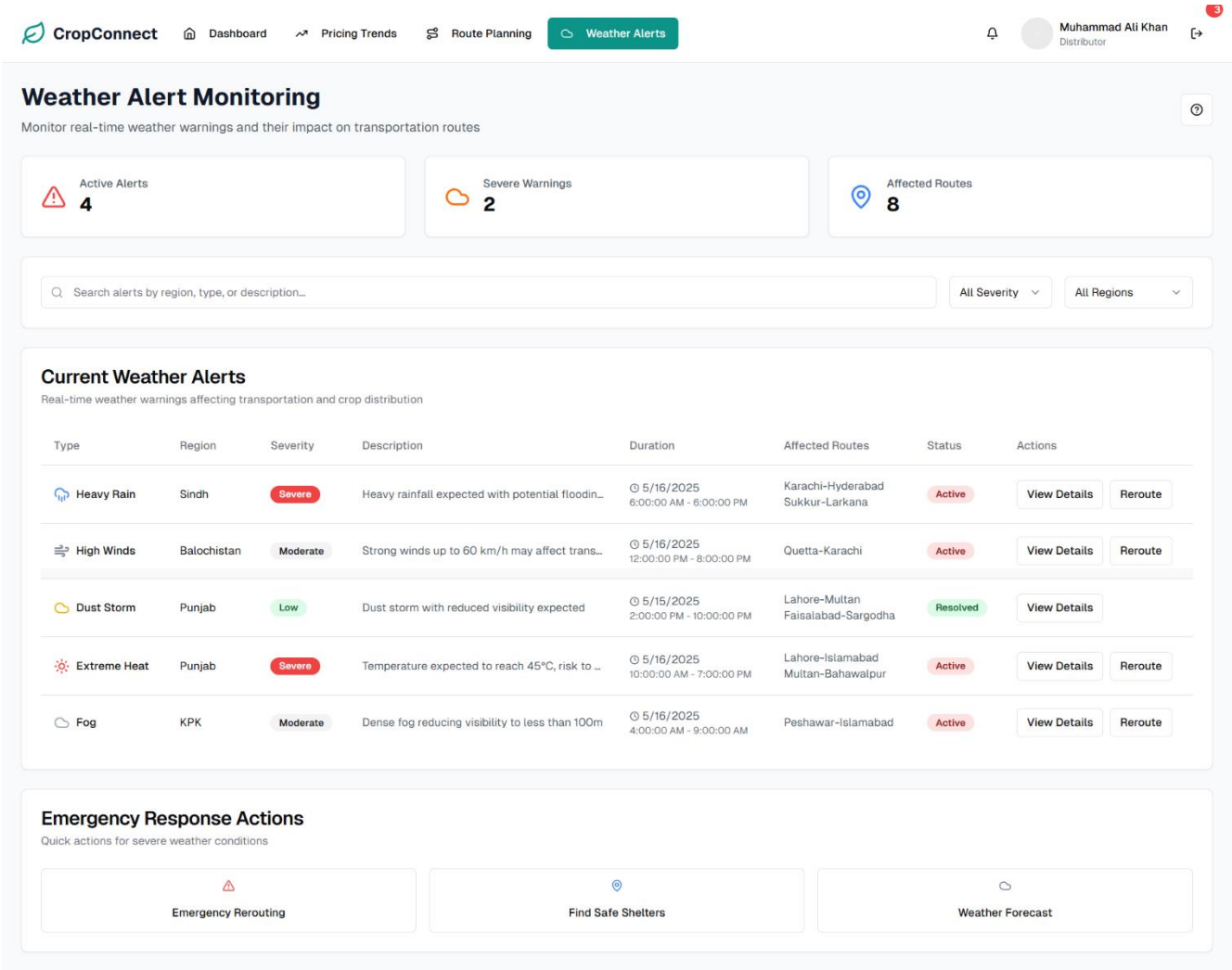


NGO Surplus/deficit Analysis





Route Planning & Reports



Weather Alerts

6.2 Software interfaces

This section lists all external and internal software components with which CropConnect interacts. Each entry describes the component and the system's high-level integration requirements, reflecting a modular (not microservice) architecture.

SI-1: Pakistan Bureau of Statistics (PBS) Data Ingestion

- **SI-1.1:** The system shall ingest crop production and price data exported from PBS in CSV or Excel format.
- **SI-1.2:** The system shall run a scheduled scraper to fetch updated PBS tables if manual files are not provided.
- **SI-1.3:** Ingested data shall be validated, mapped to internal region and crop identifiers, and stored in the production and pricing database tables.

SI-2: MSN Weather API

- **SI-2.1:** The system shall request historical weather data for each region once daily to support forecasting and trend analysis.
- **SI-2.2:** The system shall request current weather conditions hourly to detect severe weather events and feed the alerting module.
- **SI-2.3:** Received weather data and alerts shall be stored in the database and used by the Forecasting, Alerting, and Routing modules.

SI-3: Forecasting Service (Machine Learning Model)

- **SI-3.1:** The Forecasting module shall consume region, crop, and season inputs to invoke a machine learning model and generate weekly yield predictions with confidence intervals.
- **SI-3.2:** Forecast outputs shall be stored in the database and cached (e.g., in Redis) to avoid redundant model runs.

SI-4: Routing & Optimization Module

- **SI-4.1:** The Routing module shall accept source and destination region identifiers along with optimization criteria (e.g., minimize cost, avoid weather).
- **SI-4.2:** The Routing module shall calculate and return an optimized route (distance, cost estimates, and path geometry) for map rendering and analysis.

SI-5: Email Notification Component

- **SI-5.1:** The Notification module shall use an SMTP provider to send transactional emails (account verification, password resets).
- **SI-5.2:** The same module shall deliver scheduled reports and real-time alert notifications via email to stakeholders, attaching generated PDF/CSV files when required.

SI-6: In-App WebSocket Notification Component

- **SI-6.1:** Upon user login, the front end shall establish a WebSocket connection to receive real-time alerts (weather, price spikes, ingestion failures).
- **SI-6.2:** The Notification component shall push JSON-formatted alert messages over WebSocket to all connected clients for immediate display.

SI-7: Authentication & Authorization Module

- **SI-7.1:** The system shall use an OAuth 2.0/OpenID Connect provider to handle user login, role assignment, and token issuance.
- **SI-7.2:** All API calls shall require a valid JWT access token, and the Authorization module shall enforce role-based access control based on token claims.

SI-9: Redis Cache

- **SI-9.1:** The system shall cache forecast results (keyed by user, season, and parameters) in Redis for up to 24 hours to avoid repeated model runs.
- **SI-9.2:** The system shall also use Redis for short-lived session data and temporary filter states.

SI-10: Internal Module Integration

- **SI-10.1:** The application follows a modular architecture: distinct modules (Data Ingestion, Weather, Forecasting, Routing, Alerting, Authentication, Notification) reside within a single codebase and communicate via well-defined interfaces (function calls or shared libraries).
- **SI-10.2:** Shared utility components (e.g., database access layer, cache client, common data models) shall be placed in a common module and imported as needed.
- **SI-10.3:** Inter-module data exchange shall occur through in-memory objects or via database/cache lookups, avoiding REST calls between modules.

6.3 Hardware interfaces

CropConnect is a web-based platform. The following hardware interface requirements ensure compatibility with end-user devices and network infrastructure:

HI-1: Client Device & Display

- **HI-1.1:** The application will be web based responsive application which will work on different devices such as Desktop , tablets , mobile phones etc.
- **HI-1.2:** All UI components (charts, maps, tables, forms) shall adapt fluidly to viewport widths between 768 px (tablet) and 1920 px (standard desktop), without horizontal scrolling.

- **HI-1.3:** For tablet touchscreens (e.g., iPad, Android tablets), interactive elements (buttons, dropdowns, map controls) shall respond accurately to tap, swipe, and pinch gestures.

HI-2: Network Adapter & Connectivity

- **HI-2.1:** Client devices shall use standard Ethernet or Wi-Fi (802.11b/g/n/ac) to connect to the internet. A minimum bandwidth of 2 Mbps is recommended to load interactive dashboards and map tiles smoothly.
- **HI-2.2:** The system shall detect poor network conditions (e.g., latency > 200 ms or bandwidth < 500 Kbps) and automatically switch to a low-bandwidth mode—rendering charts as static images and reducing map tile resolution.

HI-3: Printing & Report Export

- **HI-3.1:** The application shall support printing of reports and dashboards via the client's default operating system print dialog, targeting hardware printers configured on Windows, macOS, and Linux.
- **HI-3.2:** Generated PDF exports (e.g., production reports, price trend charts) shall be optimized for standard A4 paper (8.27 × 11.69 inches) with 1-inch margins, ensuring compatibility with any A4-capable printer.

6.4 Communications Interfaces

This section specifies the communication functions that CropConnect will utilize, including email, web browser interactions, network protocols, and other forms of electronic communication.

CI-1: Email Notification System

- **CI-1.1:** The application shall send email notifications to users for account registration confirmations, password resets, and alert subscriptions (e.g., severe weather, price spikes).
- **CI-1.2:** Emails shall be sent through a secure SMTP service, ensuring TLS encryption.

CI-2: Scheduled Report Delivery

- **CI-2.1:** Users shall be able to schedule periodic reports (daily, weekly, or monthly), which the system will compile and send via email in PDF or CSV format.
- **CI-2.2:** Report emails shall include a brief summary and a secure link for downloading the full report.

CI-3: Web API Communication

- **CI-3.1:** The application shall communicate with external APIs (PBS ingestion endpoints, MSN Weather API) over HTTPS to ensure secure data exchange.
- **CI-3.2:** All internal RESTful interactions (e.g., front-end to back-end) shall use JSON over HTTPS.

CI-4: Real-Time Alerts (WebSocket)

- **CI-4.1:** The front end shall establish a WebSocket connection upon login to receive real-time alerts (weather warnings, price spikes, data ingestion failures).
- **CI-4.2:** Users shall see live notifications in the UI and have the option to dismiss or acknowledge them.

CI-5: Browser Interaction

- **CI-5.1:** Users shall have the option to open external links (e.g., news articles, documentation) in their device's default web browser.
- **CI-5.2:** All links opened by the application shall use HTTPS to ensure secure browsing.

CI-6: Network Protocols

- **CI-6.1:** The application shall rely on standard network protocols—TCP/IP for basic connectivity and HTTPS for all data transfers.
- **CI-6.2:** The system shall detect and adapt to low-latency or limited-bandwidth conditions by degrading visual fidelity (e.g., static chart images).

7.0 References

List any documents or other resources to which this SRS refers, if any. These might include user interface style guides, standards, system requirements specifications, interface specifications, or the SRS for a related product. The following are a few examples of different resources.

Book
Author(s). Book title. Location: Publishing company, year, pp.
Example:
W.K. Chen. Linear Networks and Systems. Belmont, CA: Wadsworth, 1993, pp. 123-35.

Article in a Journal
Author(s). "Article title". Journal title, vol., pp, date.
Example:
G. Pevere. "Infrared Nation." The International Journal of Infrared Design, vol. 33, pp. 56-99, Jan. 1979.

Articles from Conference Proceedings (published)
Author(s). "Article title." Conference proceedings, year, pp.
Example:
D.B. Payne and H.G. Gunhold. "Digital sundials and broadband technology," in Proc. IOOC-ECOC, 1986, pp. 557-998.

World Wide Web
Author(s)*. "Title." Internet: complete URL, date updated* [date accessed].
M. Duncan. "Engineering Concepts on Ice. Internet: www.iceengg.edu/staff.html, Oct. 25, 2000 [Nov. 29, 2003].

Appendix A

Example: Context Diagram

The Cafeteria Ordering System is a new software system that replaces the current manual and telephone processes for ordering and picking up meals in the Process Impact cafeteria. The context diagram in Figure A-1 illustrates the external entities and system interfaces.

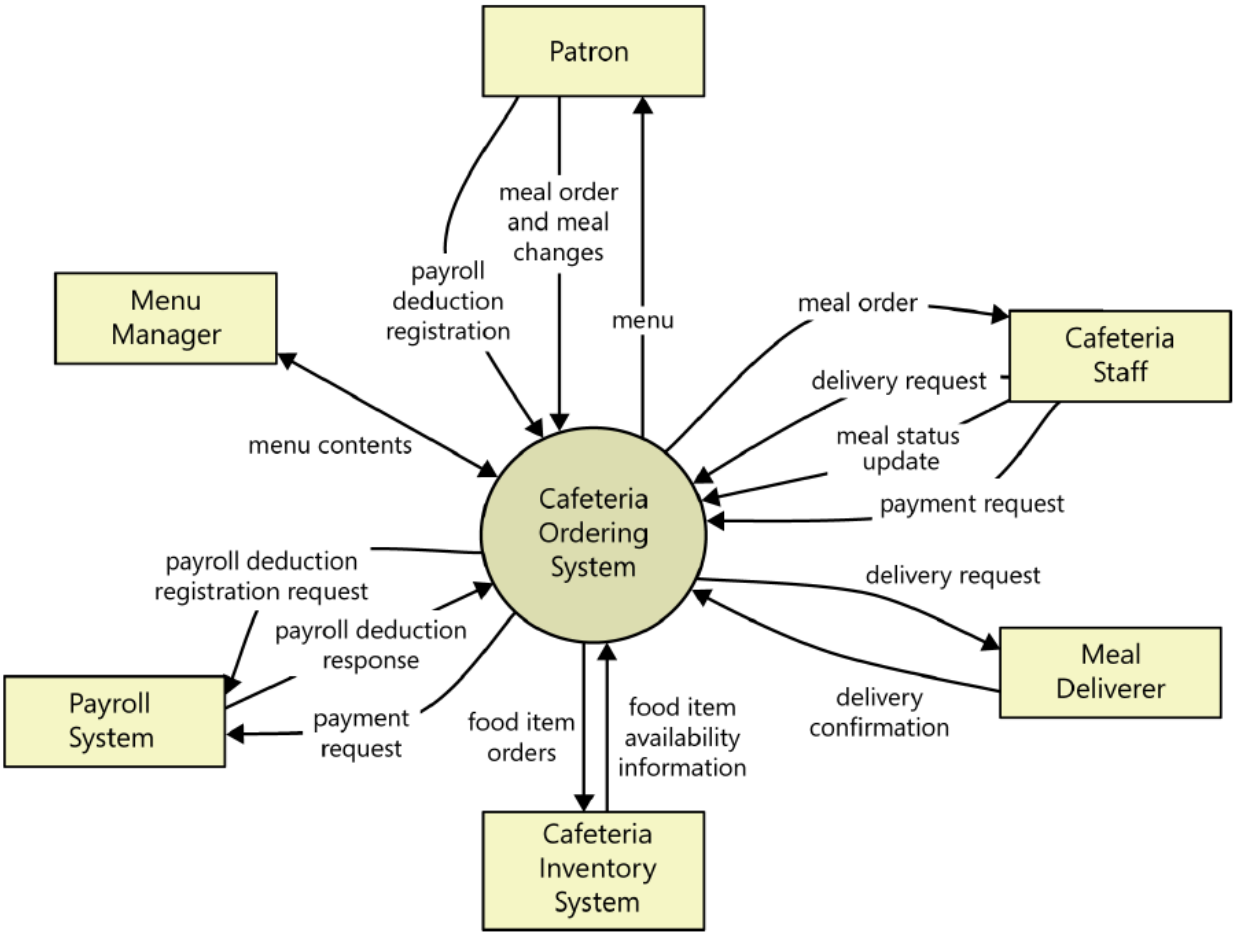


Figure A-1: Context diagram of the Cafeteria Ordering System.

Example: User Classes and Characteristics

Table A-1 Shows user classes and characteristic for Cafetaria ordering system

User class	Description
Patron	A Patron is a Process Impact employee who wants to order meals to be delivered from the company cafeteria. There are about 600 potential Patrons, of which 300 are expected to use the COS an average of 5 times per week each. Patrons will sometimes order multiple meals for group events or guests. An estimated 60 percent of orders will be placed using the corporate intranet, with 40 percent of orders being placed from home or by smartphone or tablet apps.
Cafeteria Staff	The Process Impact cafeteria employs about 20 Cafeteria Staff who will receive orders from the COS, prepare meals, package them for delivery, and request delivery. Most of the Cafeteria Staff will need training in the use of the hardware and software for the COS.
Menu Manager	The Menu Manager is a cafeteria employee who establishes and maintains daily menus of the food items available from the cafeteria. Some menu items may not be available for delivery. The Menu Manager will also define the cafeteria's daily specials. The Menu Manager will need to edit existing menus periodically.
Meal Deliverer	As the Cafeteria Staff prepare orders for delivery, they will issue delivery requests to a Meal Deliverer's smartphone. The Meal Deliverer will pick up the food and deliver it to the Patron. A Meal Deliverer's other interactions with the COS will be to confirm that a meal was (or was not) delivered.

Example: Use case Diagram

Following use case diagram is of an appointment system in which all the use case diagram relationships are presented. In further in **Table A-2** to the detail of use case diagram sytax is provided.

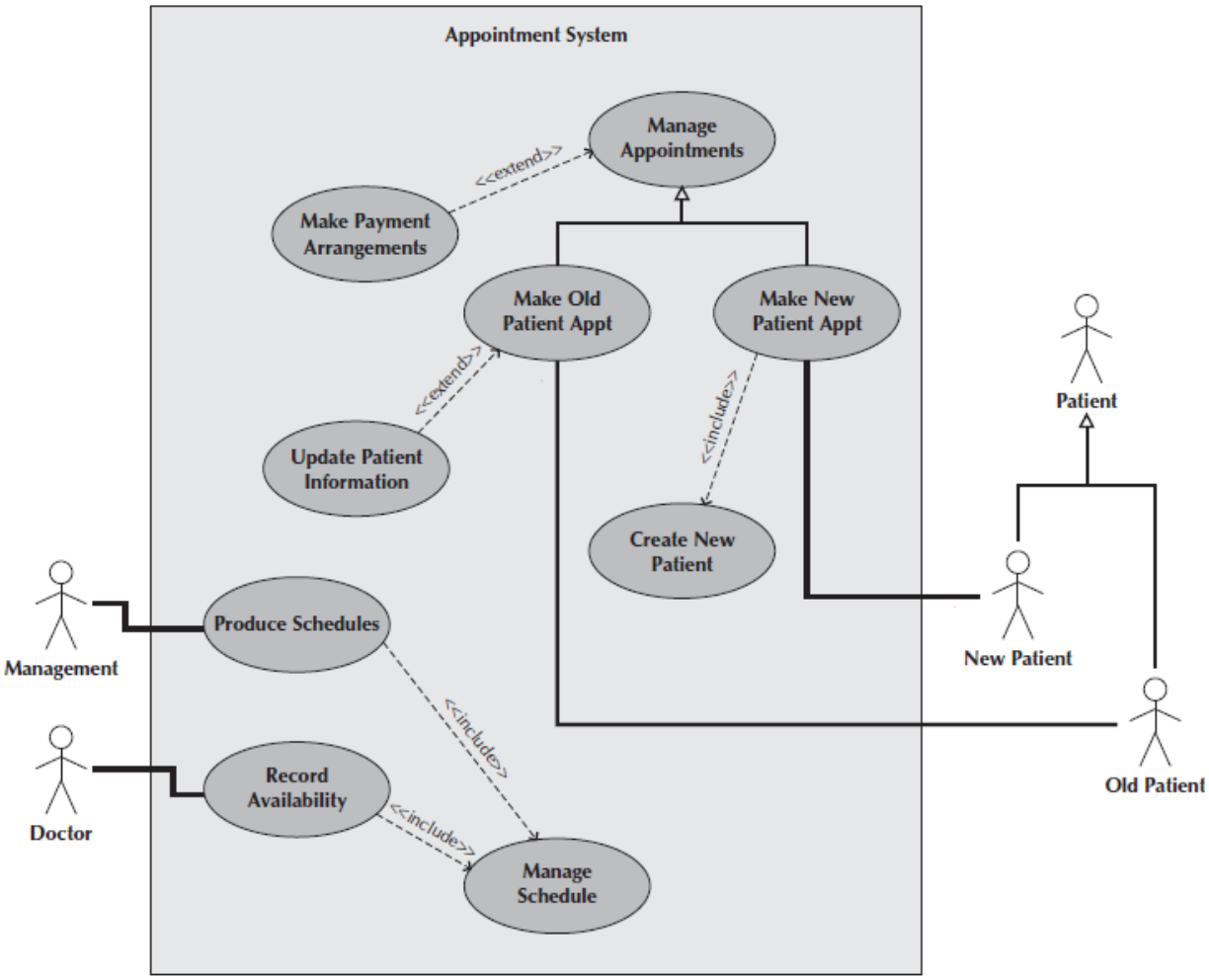
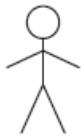
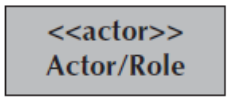
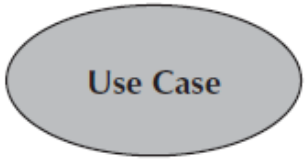
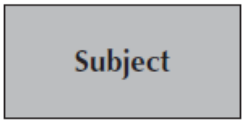
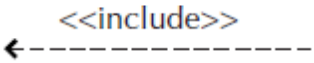
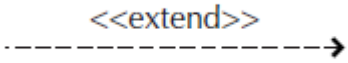


Figure A-2: Use Case Diagram of an Appointment System

Syntax for Use case Diagram

Table A-2 Syntax for Use case Diagram

An actor: ■ Is a person or system that derives benefit from and is external to the subject. ■ Is depicted as either a stick figure (default) or, if a nonhuman actor is involved, a rectangle with <<actor>> in it (alternative). ■ Is labeled with its role. ■ Can be associated with other actors using a specialization/superclass association, denoted by an arrow with a hollow arrowhead. ■ Is placed outside the subject boundary.	 Actor/Role 
A use case: ■ Represents a major piece of system functionality. ■ Can extend another use case. ■ Can include another use case. ■ Is placed inside the system boundary. ■ Is labeled with a descriptive verb–noun phrase.	
subject boundary: ■ Includes the name of the subject inside or on top. ■ Represents the scope of the subject, e.g., a system or an individual business process.	
An association relationship: ■ Links an actor with the use case(s) with which it interacts.	
An include relationship: ■ Represents the inclusion of the functionality of one use case within another. ■ Has an arrow drawn from the base use case to the used use case.	
An extend relationship: ■ Represents the extension of the use case to include optional behavior. ■ Has an arrow drawn from the extension use case to the base use case.	
A generalization relationship: ■ Represents a specialized use case to a more generalized one. ■ Has an arrow drawn from the specialized use case to the base use case.	

Detail Use Case Example

The **Table A-3** below indicate a comprehensive use case template filled in with an example drawn from the Cafeteria ordering system (COS).

Table A-3 Show the detail use case template and example

Use Case ID:	Enter a unique numeric identifier for the Use Case. e.g. UC-1
Use Case Name:	Enter a short name for the Use Case using an active verb phrase. e.g. Order a Meal
Actors:	[An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks.] e.g. Primary Actor: Patron Secondary Actors: Cafeteria Inventory System
Description:	[Provide a brief description of the reason for and outcome of this use case.] e.g. A Patron accesses the Cafeteria Ordering System from either the corporate intranet or external Internet, views the menu for a specific date, selects food items, and places an order for a meal to be picked up in the cafeteria or delivered to a specified location within a specified 15-minute time window.
Trigger:	[Identify the event that initiates the use case.]e.g. A Patron indicates that he wants to order a meal.
Preconditions:	[List any activities that must take place, or any conditions that must be true, before the use case can be started. PRE-1. Patron is logged into COS. PRE-2. Patron is registered for meal payments by payroll deduction.
Postconditions:	[Describe the state of the system at the conclusion of the use case execution. POST-1. Meal order is stored in COS with a status of "Accepted." POST-2. Inventory of available food items is updated to reflect items in this order. POST-3. Remaining delivery capacity for the requested time window is updated.
Normal Flow:	[Provide a detailed description of the user actions and system responses that will take place during execution of the use case under normal, expected conditions. 1.0 Order a Single Meal 1. Patron asks to view menu for a specific date. (see 1.0. E1, 1.0.E2) 2. COS displays menu of available food items and the daily special. 3. Patron selects one or more food items from menu. (see 1.1) 4. Patron indicates that meal order is complete. (see 1.2) 5. COS displays ordered menu items, individual prices, and total price, including taxes and delivery charge. 6. Patron either confirms meal order (continue normal flow) or requests to modify meal order (return to step 2). 7. COS displays available delivery times for the delivery date. 8. Patron selects a delivery time and specifies the delivery location. 9. Patron specifies payment method. 10. COS confirms acceptance of the order. 11. COS sends Patron an email message confirming order details, price, and delivery instructions. 12. COS stores order, sends food item information to Cafeteria Inventory System, and updates available delivery times.
Alternative Flows:	[Document legitimate branches from the main flow to handle special conditions (also known as extensions). For each alternative flow reference the branching step number of the normal flow and the condition which must be true for this extension to be executed. e.g. 1.1 Order multiple identical meals 1. Patron requests a specified number of identical meals. (see 1.1. E1) 2. Return to step 4 of normal flow. 1.2 Order multiple meals 1. Patron asks to order another meal. 2. Return to step 1 of normal flow. Note: Insert a new row for each distinctive alternative flow.]
Exceptions:	1.0. E1 Requested date is today and current time is after today's order cutoff time 1. COS informs Patron that it's too late to place an order for today. 2a. If Patron cancels the meal ordering process, then COS terminates use case.

	2b. Else if Patron requests another date, then COS restarts use case. 1.0. E2 No delivery times left 1. COS informs Patron that no delivery times are available for the meal date. 2a. If Patron cancels the meal ordering process, then COS terminates use case. 2b. Else if Patron requests to pick the order up at the cafeteria, then continue with normal flow, but skip steps 7 and 8. 1.1. E1 Insufficient inventory to fulfill multiple meal order 1. COS informs Patron of the maximum number of identical meals he can order, based on current available inventory. 2a. If Patron modifies number of meals ordered, then return to step 4 of normal flow. 2b. Else if Patron cancels the meal ordering process, then COS terminates use case.
Business Rules	Use cases and business rules are intertwined. Some business rules constrain which roles can perform all or parts of a use case. Perhaps only users who have certain privilege levels can perform specific alternative flows. That is, the rule might impose preconditions that the system must test before letting the user proceed. Business rules can influence specific steps in the normal flow by defining valid input values or dictating how computations are to be performed e.g. BR-1 Delivery time windows are 15 minutes, beginning on each quarter hour. BR-2 Deliveries must be completed between 11:00 A.M. and 2:00 P.M. local time, inclusive. Note: If you are maintaining the business rule in a separate table in SRS then only mention here their IDs.
Assumptions:	[List any assumptions. 1. e.g. Assume that 15 percent of Patrons will order the daily special (Source: previous 6 months of cafeteria data).

Event-Response Tables

In order to develop Event Response table first it is required to identify all possible events. Following is an example of events list and event-response table of highway intersection system.

The highway intersection system described earlier has to deal with various events, including these:

- A sensor detects a car approaching in one of the through lanes.
- A sensor detects a car approaching in a left-turn lane.
- A pedestrian presses a button to request to cross a street.
- One of many timers counts down to zero.

Table A-4 presents a fragment of what an event-response table might look like for such a system. Each expected system behavior consists of a combination of event, system state, and response.

Note: You may add more information to event-response table in order to perform detail requirement analysis which includes:

- The event frequency (how many times the event takes place in a given time period, or a limit to how many times it can occur).
- Data elements that are needed to process the event.
- The state of the system after the event responses are executed.

Table A-4 Partial Event-Response Table for a Highway Intersection

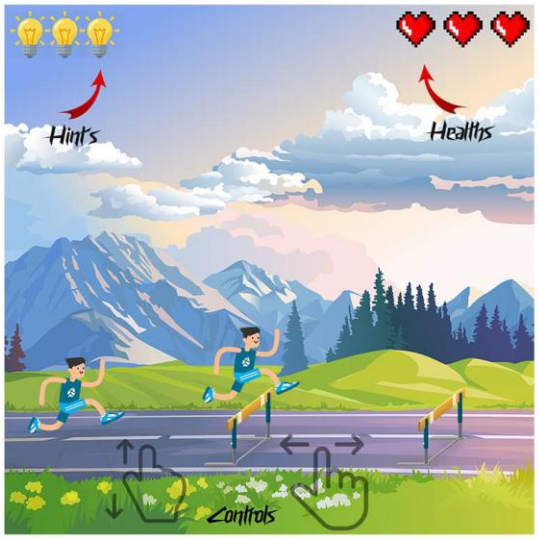
Event	System State	Response
Road sensor detects vehicle entering left-turn lane.	Left-turn signal is red. Cross-traffic signal is green.	Start green-to-amber countdown timer for cross-traffic signal.
Green-to-amber countdown timer reaches zero.	Cross-traffic signal is green.	1. Turn cross-traffic signal amber. 2. Start amber-to-red countdown timer.
Amber-to-red countdown timer reaches zero.	Cross-traffic signal is amber.	1. Turn cross-traffic signal red. 2. Wait 1 second. 3. Turn left-turn signal green. 4. Start left-turn-signal countdown timer.
Pedestrian presses a specific walk-request button.	Pedestrian sign is solid Don't Walk. Walk-request countdown timer is not activated.	Start walk-request countdown timer.
Pedestrian presses walk-request button.	Pedestrian sign is solid Don't Walk. Walk-request countdown timer is activated.	Do nothing.
Walk-request countdown timer reaches zero plus the amber display time.	Traffic signal in walk direction is green.	Change all green traffic signals to amber.
Walk-request countdown timer reaches zero.	Traffic signal in walk direction is amber.	1. Change all amber traffic signals to red. 2. Wait 1 second. 3. Set pedestrian sign to Walk. 4. Start don't-walk countdown timer.

Story Boarding

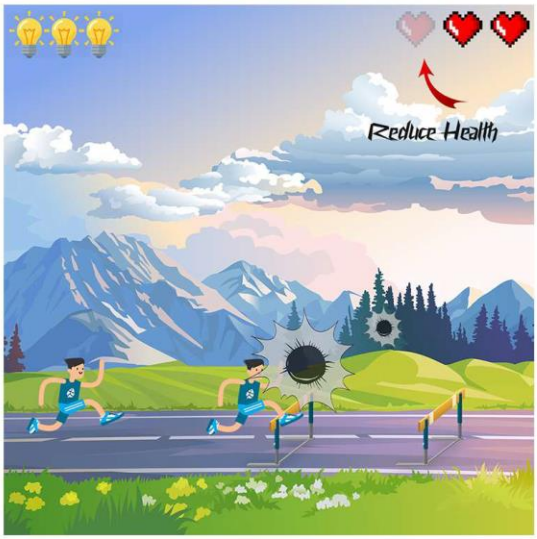
Following is an example of story boarding which is use to identify and analyze graphically intensive applications.

1- The main intro animation text of the game will be played

2- The main menu will appear once all the assets are load. The player can start the game, see the settings and leaderboard.

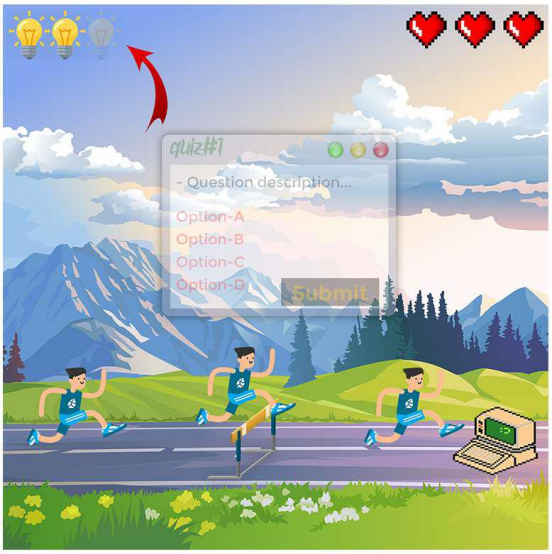


3- Gameplay consists of a running character and hurdles on the way, the player movement will be controlled using swipes on the screen.



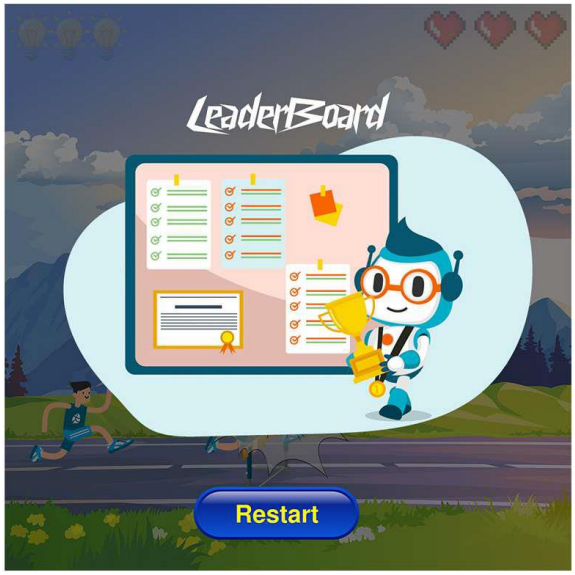
4- If the player collides with a static hurdle, the main health will be reduced.

5- There will be a quiz system that will pop up once the player hits a special hurdle in the game.



6- If the player failed to solve the quiz, the hint power will be reduced, all hint powers gone means one health reduced.

6- Once there are no healths remaining and the player hits an hurdle, the dying animation will be played and the game will be over.



7- The leaderboard will be shown at the end that will provide the summary of the quizzes attempted.